FORESIGHT

Network Attack Forecasting
Technical Architecture

SIH26153 | NTRO | Software
Deja Vu T-1
9 September 2026

An implementation-grade handbook for evidence-accountable forecasting, learned world-state rollouts and policy-constrained predictive defense.

Build the core. Test every claim. Promote research by evidence.

Includes canonical contracts, model dimensions, training objectives, falsification gates, a six-person delivery plan, ten research candidates and a staged startup roadmap.

Design specification, not a report of trained-model results. The separate two-page summary is the SIH architecture submission artifact.

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1. Authority and Build Decision

FORESIGHT is an offline-first, evidence-accountable network attack forecasting system, with a staged path to a policy-constrained predictive defense platform. This document is the engineering authority for the next implementation. It supersedes the supplied integrated v4 architecture; no earlier configuration, equation, pseudocode, or diagram overrides it. Historical decisions are summarized in Appendix A, not retained as competing implementation instructions.

Build decision: proceed with the contracts, data qualification, replay harness and Tier A reference implementation. Do not interpret this design as trained-model validation, production certification, a guarantee of unknown-exploit prevention, or a guarantee of winning SIH. Acceptance requires the experiments and release gates specified here. Advanced components are implemented as isolated candidates and promoted only by evidence.

The complete source reviewed was `FORESIGHT_FINAL_Integrated_Architecture_v4.pdf`, 87 pages, SHA-256 `66f21e754b4f627bdeb6de2d3b49e91f7e74021859e6c77a6e9193fbab64e969`. Its strengths are preserved: forecasting rather than static classification, graph-temporal context, separate novelty, offline operation, provenance, sealed replay, measured explanations and safe control boundaries. Its conflicting dimensions, labels, probability semantics and action claims are replaced.

The product name is **FORESIGHT**. The machine-readable contract identifier is `foresight.core.2026-09`; this is not a marketing version. The SIH team is Deja Vu T-1. No team ID is included. The product and research roadmap deliberately exceed the competition scope, but the competition does not depend on those extensions.

Reading and Implementation Rules

- Sections 2-30 and 35 contain the normative platform and delivery contracts. MUST indicates a release requirement; SHOULD permits a documented exception; MAY identifies an optional implementation.

- Sections 31-34 contain the research tournament, original-candidate register, adversarial review and startup roadmap. A research hypothesis is not a production requirement or a measured result.

- Appendix A records the architecture autopsy and resolved decisions. Appendix B is the reference register. Numerical budgets are engineering acceptance targets, never achieved results.

- The accompanying two-page PDF is the submission architecture summary. This longer handbook is for the engineering team, not a replacement for the official two-page submission limit.

2. SIH26153 Compliance

The official SIH portal entry was read directly on 9 September 2026. The statement is NTRO's **AI based Network Attack Forecasting from Network Traffic Data**, software, Blockchain & Cybersecurity. It explicitly permits alternative architectures while requiring learned transition dynamics, forward forecasting, interpretable outputs and both traffic-feature levels. Authentication and other public telemetry are permitted additions, not replacements for traffic requirements. [1]

Requirement	FORESIGHT implementation	Required evidence
Structured network state	Timestamped directed host graph, flow/packet features, validity masks and scope registry	Feature fixtures, graph snapshots and channel coverage report
Learned state transitions	Entity-memory and global-temporal model; explicit prior transition and future-observation decoders	Held-out observable prediction versus persistence; free-rollout error
Supervised timeline learning	Event labels and stage tags derived from documented attack timelines; observable next-state targets from subsequent telemetry	Event manifest, label tiers, masked losses and timeline alignment
K-step forward simulation	Roll the learned state forward without future telemetry; hazards and stage evidence read from imagined states	Sealed forecast and per-step transition trace
Infiltration probability timeline	Precisely named endpoint/event cumulative probabilities, including a supported intrusion-onset or progression target	Event-matched metrics and calibration by horizon
ATT&CK mapping	Multi-label tactic evidence; requested five-tactic projection for SIH	Versioned mapping, confidence tier and confusion report
Explainability	Traffic feature groups, source flows, hosts and temporal edges; deletion-tested attribution	Evidence IDs and fidelity versus random deletion
PCAP or CSV interface	Local file ingestion, capability report, offline forecast timeline and flagged-flow evidence	Fresh-machine offline demonstration
Logistic regression benchmark	Same legal features, targets, split, scopes and operating budget	F1, precision, recall, FPR; additional operational metrics
Open-source reproducibility	Source, training scripts, weights, dependency lock, data-fetch instructions and licenses	Clean rebuild and reproduction manifest

The official evaluation package lists a source link, setup README, architecture document of at most two pages, video of at most two minutes, and presentation of at most five slides. These are total artifact limits unless the organizer explicitly grants a different rule. The previous six-slide deck must not be assumed compliant merely because a local template contained an extra slide. [1]

Interpretation boundary: the statement uses causal-simulation language. A model of observational transitions does not by itself identify the causal effect of isolation or account revocation. FORESIGHT satisfies forward simulation with a learned predictive world model and reserves intervention claims for separately identified experiments. [14]

3. Mission and Information Boundaries

The governing question is: **given the evidence actually available now, which measurable harmful transition is becoming plausible, how much useful decision time remains, and what can the defender safely do with that information?** The architecture distinguishes three products rather than forcing all of them into one risk number.

Product	Estimand	Evidence needed	Authority
Observational forecast	Future event under the recorded/current operating policy	Causal prefix, event targets, observed future	Inform and prioritize
Security-state belief	Distribution over hidden compromise hypotheses	Explicit observation model, prior assumptions, multi-sensor evidence	Investigate; hypotheses remain labeled
Intervention estimate	Outcome distribution under a specified action and policy	Controlled interventions or justified identification assumptions	Recommend; execute only through policy

The full loop is **observe -> maintain evidence and belief -> forecast -> evaluate decision options -> request evidence or propose action -> authorize -> execute -> verify -> adjudicate -> learn in shadow**. Requesting evidence, waiting, escalating and doing nothing are explicit actions. Containment is not an inevitable final stage. Learning never happens automatically in the same process that issues controls.

An observation is not automatically true because it is signed. A latent vector is not a discovered causal state. A novel event is not automatically malicious. A feasible attack path is not an attack that will occur. A lower post-action model score is not proof of prevented compromise. These distinctions MUST survive serialization, the UI, reports and marketing.

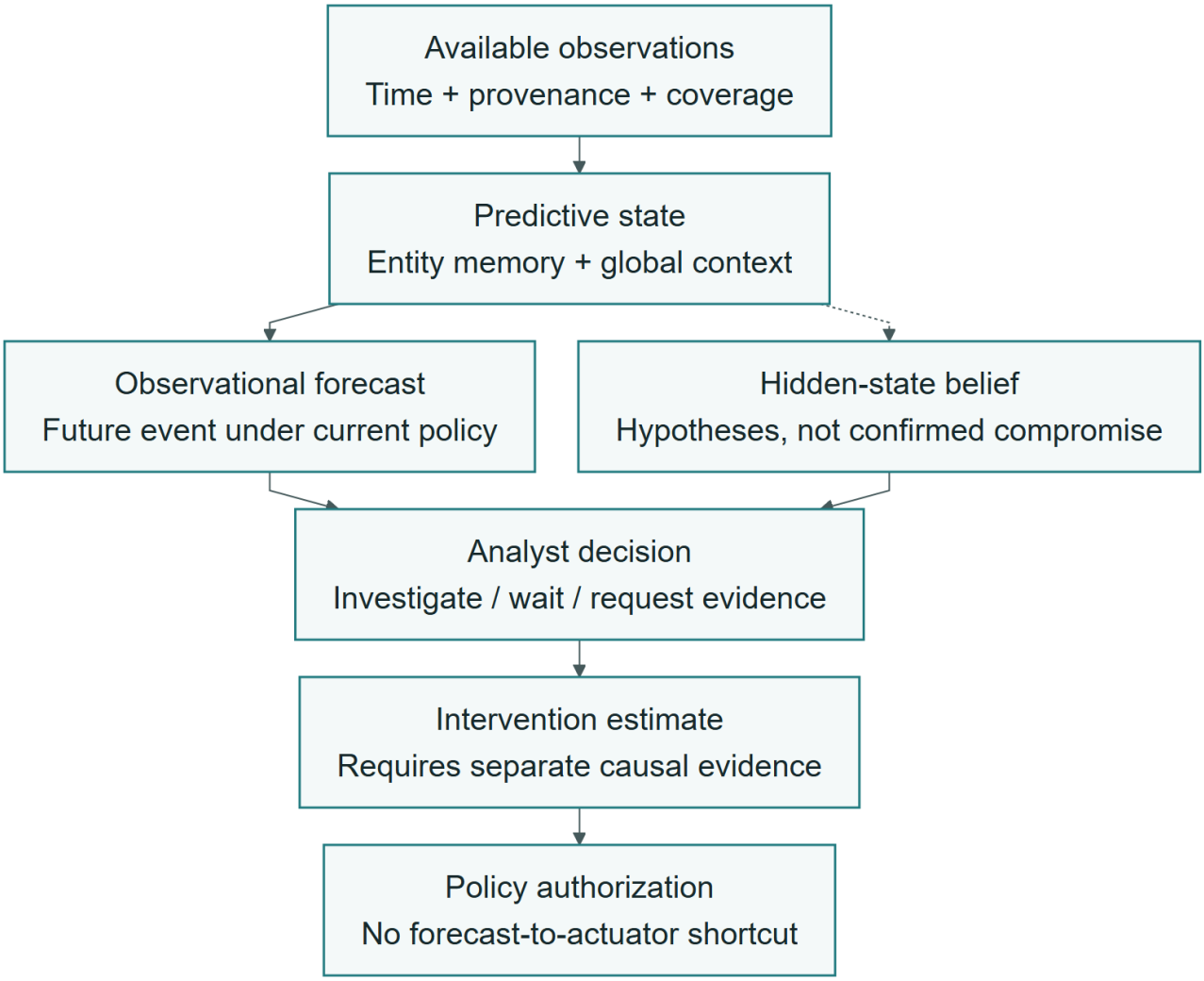


Figure 1. Forecasts, hidden-state hypotheses and intervention estimates have different evidence and authority requirements.

4. Delivery Tiers

Tier A: SIH Core

Seven deliverables define the minimum strong system: (1) qualified flow-plus-packet data and causal replay; (2) per-entity graph-temporal forecasting with a corrected GAT/LSTM reference; (3) observable multi-step world-state prediction; (4) event/stage forecasts with calibrated horizon semantics; (5) SENTINEL novelty and explicit abstention; (6) fidelity-tested, source-linked explanations; (7) sealed offline demo and a common baseline/evaluation harness.

Tier A includes per-entity recurrent memory, but not a full Bayesian enterprise simulator. It predicts candidate future communication edges as a world-model output. It MUST NOT label an unvalidated chain of those edges a calibrated attack-path probability. It supports a network-only judged configuration without requiring endpoint agents or customer integrations.

Tier B: High-Value Stretch

In order: one endpoint or identity adapter; next-entity/typed-event decoder and path ranking; one targeted evidence request; an independent three-model ensemble; one authorized range action with a paired no-action continuation. Each is separately feature-flagged. A failed stretch gate cannot remove a working core capability.

Tier C: Startup and Research

Continuous-time heterogeneous memory, observation-conditioned belief distributions, cost-aware active sensing, joint path distributions, causal action models, adaptive-attacker scenarios, controlled learning, multi-tenant streaming and production response adapters. These are architectural interfaces and research protocols now, not promises that every component is ready to deploy.

5. Component Hierarchy

Layer	Components	Persisted artifact
Observation	CSV/PCAP adapters; endpoint/identity/cloud adapters; hostile-file sandbox	Immutable raw object, parser report
Truth and time	Clock policy, source lineage, deduplication, observation availability, label separation	Event and feature records
State	Scope registry, probabilistic identity links, directed graph, entity memory, global context	Versioned state checkpoint
Prediction	Observable decoder, transition prior, hazard readout, tactic tags, candidate edges	ForecastRecord
Trust	Calibration, selective prediction, novelty, drift diagnosis, coverage accounting	TrustContract
Evidence	Grouped attribution, temporal/edge deletion, flow provenance	EvidenceItem
Decision	Query selector, path engine, action simulator, policy evaluation	QueryProposal / ActionProposal
Control	Approval, bounded executor, postconditions, expiry reconciler, recovery	ActionOutcome
Learning	Adjudication, quarantine, reference model, shadow candidate, promotion	TrainingManifest / ModelCard
Product	Offline UI, CLI, evidence export; later tenant APIs and analyst integration	Reproduction bundle

Core processes are the local API/UI, a resource-limited parser worker, the inference worker and an evaluator/reveal process. Inference has no read access to future suffixes or evaluation labels. The optional range executor is a separate process with its own policy and credentials. [20],[21]

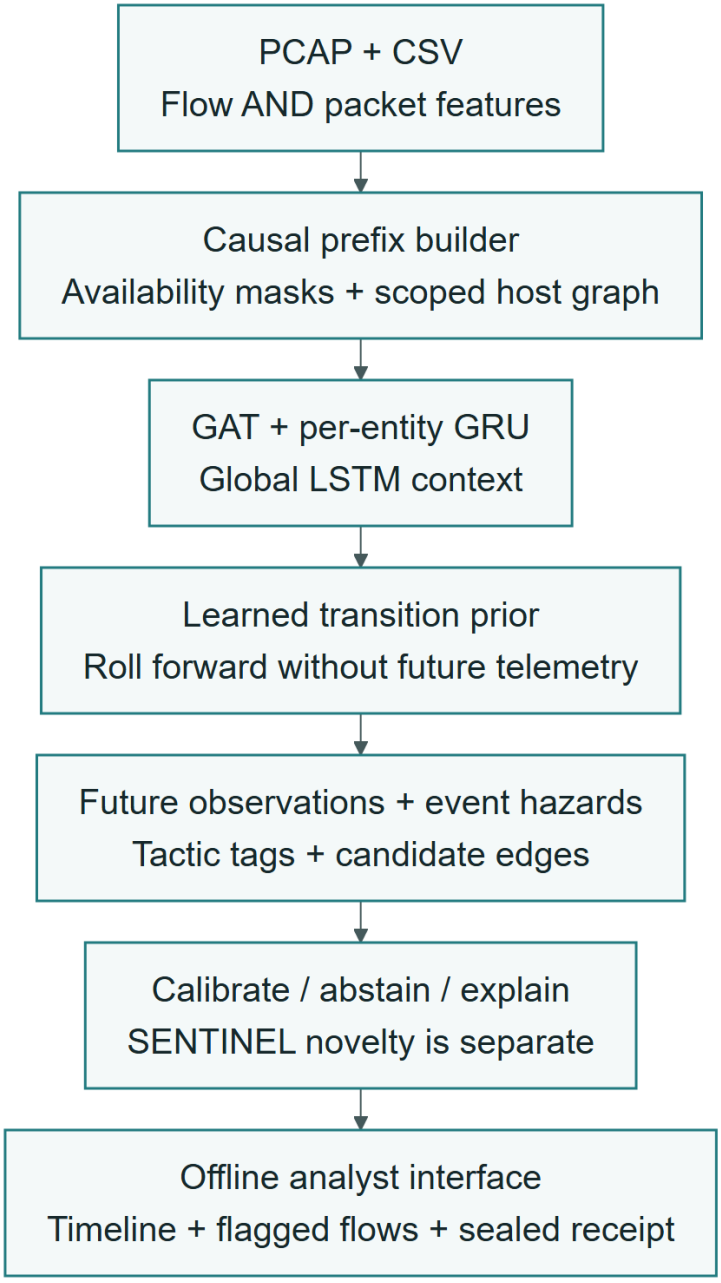


Figure 2. The judged offline pipeline. Every forecast is produced from a legal prefix and linked to source evidence.

6. Data Qualification Before Model Training

The dataset registry is an engineering gate, not a bibliography. Each release declares exact files, hashes, permission/license terms, capture topology, time zone, clock uncertainty, packet completeness, endpoint linkage and label provenance. Never infer availability of packet attributes merely because a dataset has flow CSVs. Never synthesize missing packet features and present them as observed.

Dataset	Defensible role	Qualification work	Not established by use
CSE-CIC-IDS2018	Flow/PCAP baseline; documented attack episodes	Audit extraction errors, endpoint mapping, completed-flow availability and campaign splits	Production generalization or complete compromise-stage truth
CTU-13	Botnet/normal/background distinction and transfer stress	Keep background separate from verified normal; align labels to actual capture	All background is benign; zero-day coverage
CICAPT-IIoT2024	Network plus provenance and attack-step alignment	Join attack metadata, process entities and packet times; count independent campaigns	Large independent enterprise population
LANL authentication or another public identity source	Separate identity-model experiments	License, timestamp meaning, entity scope and red-team label coverage	Synchronized multimodal ground truth with unrelated PCAP
Authorized range	Intervention and missing-sensor experiments	Fixed scenarios, randomization, service checks and isolated network	Independent external real-world proof

CSE-CIC-IDS2018 provides traffic captures, logs and derived features; CTU-13 distinguishes normal and background; CICAPT-IIoT provides PCAP and provenance/attack information. The exact files used still require local audit. Cross-dataset joins by coincident timestamps are prohibited: unrelated traces are not simultaneous observations of one enterprise. [2],[3],[4],[5]

The release manifest MUST include counts of independent episodes and confirmed target events in training, development, calibration and locked test; number of endpoints; pre-event history available; full/partial future coverage; and ambiguous labels. A task without sufficient event-level support is disabled or labeled exploratory. Millions of correlated windows are not millions of independent attacks.

First gate: reproduce one nontrivial episode from raw telemetry, show the forecast prefix and future target provenance, and demonstrate that features do not depend on the suffix. Do not spend GPU time before this passes.

7. Time, Availability and Sealing

All internal times are signed int64 Unix nanoseconds in UTC; JSON represents them as decimal strings to avoid JavaScript precision loss. Source clock uncertainty is stored as an interval. Event time describes occurrence; `available_at` describes earliest computability; `ingested_at` describes local receipt; `decision_emitted_at` describes actual publication. These are distinct.

The normative Tier A rule uses a decision cutoff `d`: only records with event time, available time and ingestion time no later than `d` may influence the forecast. Window close time can precede `d`. There is no exception that lets late arrivals enter a forecast while the output pretends it was issued earlier. A watermark may delay publication, but that delay reduces reported actionable lead time.

```
eligible(field, d) = (
    field.event_time_upper_ns <= d
    and field.available_at_ns <= d
    and field.ingested_at_ns <= d
    and field.provenance_verified
)
lead_seconds = (target_time_ns - decision_emitted_at_ns) / 1e9
actionable_lead = lead_seconds - review_delay - action_effect_delay
```

Closed-flow features become available only at flow closure plus processing delay. Partial-flow features use only packets received so far and a separate `partial_flow` flag. Statistics spanning a future packet are forbidden. Dataset-derived synthetic ingestion clocks are permitted for replay only if disclosed; resulting lead time is replay lead time, not measured live pipeline latency.

The seal covers canonical prefix content, scope, cutoff, parser/scaler/model/calibrator/policy versions, forecast bytes, random seed and output time. Use deterministic canonical JSON for metadata and a stable typed row encoding for features, sorted by `(ingested_at, sensor_id, source_sequence, event_id)`. NaNs and unordered dictionaries are prohibited in seal material. Hash the forecast itself, not only the input.

Local SHA-256 makes edits detectable relative to a retained commitment; it does not independently prove wall-clock issuance or prevent a team from preselecting an easy case. In the demo, the judge selects an eligible episode, the reveal process holds the suffix, and the commitment is displayed before reveal. Production adds authenticated append-only storage and an independently witnessed log checkpoint where required.

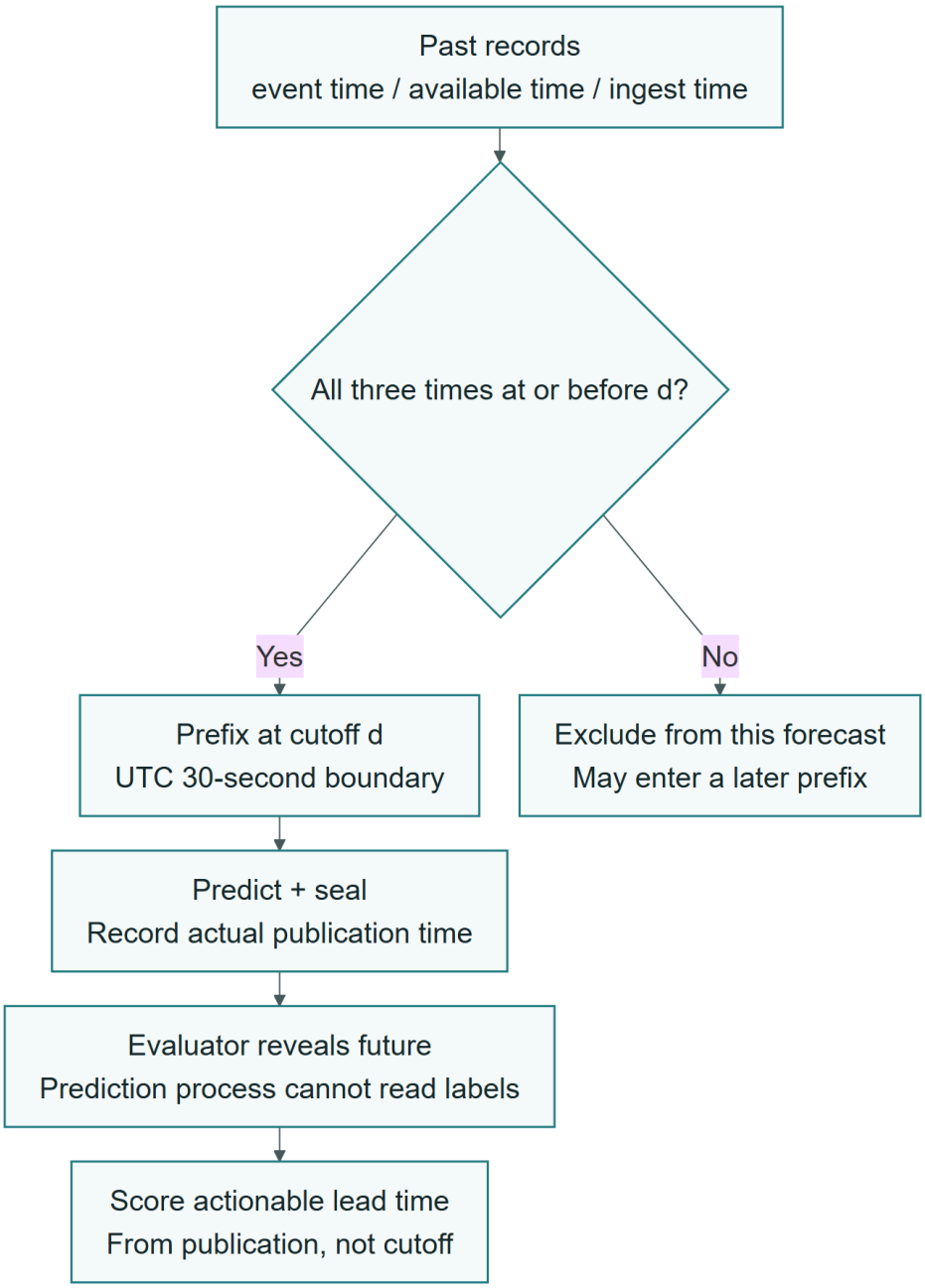


Figure 3. Availability-aware replay prevents future observations and publication delays from becoming fictitious lead time.

8. Canonical Events, Identity and Scope

```
SecurityEvent = {
  schema: "foresight.core.2026-09", tenant_id, event_id,
  sensor_id, source_sequence, source_type, parser_hash,
  event_time_lower_ns, event_time_upper_ns,
  available_at_ns, ingested_at_ns,
  subjects: [EntityRef], objects: [EntityRef], event_kind,
  observables, validity, source_event_ids, raw_object_hash,
  raw_pointer, evidence_group_id, privacy_class, retention_class
}
EntityRef = {
  tenant_id, entity_id, entity_type, identity_version,
  valid_from_ns, valid_until_ns, link_confidence, source_refs
}
```

Identity keys enable tracking but are not numeric model features. An IP address is an observed network address, not a permanent device identity. DHCP, NAT, container reuse and recycled process IDs MUST NOT silently merge histories. Prefer tenant-scoped asset IDs and signed sensor identities; retain ambiguous associations as alternative links or an unknown identity, and invalidate affected forecasts when ambiguity exceeds the task's calibrated regime.

Every forecast names its `scope_id`, target entities, observed population, external boundary, missing sensors and truncated-neighborhood count. Tier A uses predeclared cohorts and variable-size graphs with a default cap of 128 nodes; a 32-host demo is a workload fixture, not a coverage assertion. Cohort assignment is based on inventory or past-only topology, never on attack labels. Exceeding the cap causes explicit partitioning or scope abstention, not silent top-volume dropping.

Tier B uses target-rooted neighborhoods with a reserved root and bounded halo. Production shards state by tenant and stable entity ownership, with cross-shard references and freshness bounds. A single giant graph is not the serving unit. Cold-state eviction preserves a summary, eviction reason and resulting coverage loss. External/unresolved entities retain aggregate communication context but do not receive falsely specific next-victim predictions.

9. Flow and Packet Semantics

PCAP parsing runs in a sandbox with read-only input, bounded memory, file size, decompression ratio and execution time. A native packet extractor is appropriate; Scapy remains useful for fixtures. Both flow and packet adapters produce the same canonical event semantics. The software MUST NOT run a shell command composed from an uploaded filename.

A flow is a directional five-tuple segment with a configured idle timeout and activity interval. Bidirectional relationships are derived by matching reversed tuples inside the same scope and observation interval. The defaults are a 30-second idle timeout and a 120-second active segment timeout; segment closure is an observation event, not a retroactive change. Parser configuration is hashed. Cross-sensor duplicates share an evidence lineage group rather than being counted as independent events.

Packet statistics are computed from the observed packet prefix. Retransmission is a bounded heuristic over sequence-number overlap for TCP flows with sufficient sequence visibility, not ground truth about loss; offload artifacts and incomplete capture invalidate the estimate. Failed-connection ratios use matured connection attempts with an explicit observation timeout, not all still-pending SYNs. Partial or asymmetric capture creates missingness, not failure labels.

Encrypted traffic retains packet length, timing and accessible handshake metadata. TLS 1.3 encrypts handshake messages after ServerHello, so passive certificate visibility is not guaranteed. QUIC, ECH, proxying and sensor placement further change visibility. Every parser advertises capabilities; the model does not invent fields. Payload decryption is a separately authorized enterprise integration, not a prerequisite for the SIH core. [22]

10. Feature Registry and Shapes

The canonical node registry contains **44 numeric values and 44 validity bits: 88 inputs**. Edges contain **11 values and 11 validity bits: 22 inputs**. Global context contains **24 values and 24 validity bits**, projected by a learned adapter to 24 dimensions. This replaces every earlier 42/8/20 feature assertion.

Node positions	Names, in order	Meaning
1-6	syn_count, ack_count, fin_count, rst_count, psh_count, urg_count	Observed outgoing TCP flags
7-9	syn_ack_ratio, rst_fraction, failed_connection_fraction	Counts with explicit denominators and maturity masks
10-13	unique_destination_count, unique_destination_port_count, destination_port_entropy, contiguous_port_coverage	Outgoing distinct counts and port-set structure
14-16	in_degree, out_degree, degree_ratio	Directed observed graph; ratio `(out+1)/(in+1)`
17-20	bytes_sent_mean, bytes_sent_p95, bytes_received_mean, bytes_received_p95	Per observed flow-segment directional volume
21-24	packets_sent_mean, packets_sent_p95, packets_received_mean, packets_received_p95	Per observed flow-segment directional packet totals
25	bidirectional_byte_ratio	`(sent+1)/(received+1)`
26-29	iat_mean, iat_variance, iat_max, iat_min	Outgoing packet inter-arrival seconds within observed window
30-32	duration_mean, duration_variance, duration_max	Observed elapsed flow-segment seconds
33-36	ttl_mean, ttl_variance, tcp_window_mean, tcp_window_variance	Outgoing IP TTL and raw 16-bit TCP advertised window
37-38	retransmission_fraction, fragmented_packet_fraction	Only when parser/capture conditions support them
39-43	payload_bin_0, payload_bin_1, payload_bin_2, payload_bin_3, payload_bin_4	Fractions at byte bins `[0,64),[64,128),[128,512),[512,1024),[1024,infinity)`
44	ordered_port_arrival_score	Fraction of adjacent distinct first-seen ports whose absolute difference is one

`contiguous_port_coverage` uses sorted distinct port adjacency; it does not encode scan order. `ordered_port_arrival_score` uses first-seen timestamp order with deterministic ties and requires at least two distinct ports. Entropy and distinct-port features provide evidence about randomized access, but do not establish that a scan is malicious. Variances use population denominators; quantiles use a fixed linear interpolation rule.

The edge order is `bytes, packets, flow_count, duration_mean, syn_count, rst_count, tcp_fraction, udp_fraction, other_fraction, directional_balance, bilateral_seen`. Directional balance is `(bytes_uv-bytes_vu)/(bytes_uv+bytes_vu+1)`; bilateral_seen requires observed packets both ways. Fractions use observed packets, not inferred flows. Self-loops used inside GAT are computational edges with explicit zero

attributes; they are not observed communication.

The global order is `flow_start_count, active_host_count, new_pair_fraction, density, degree_entropy`, six TCP-flag fractions, three protocol fractions, four directed-pair-byte statistics `(mean,p50,p95,p99)`, four directed-pair-packet statistics, `missing_packet_channel_fraction`, and `sensor_lag_p95_seconds`. This is exactly 24. Each has a validity bit; the global adapter receives 48 channels and emits 24.

Nonnegative unbounded counts, durations, variances and ratios use log1p; bounded fractions and normalized entropy remain unlogged. Fit median/IQR scaling on training only; replace zero IQR by one. Invalid values become zero after scaling with validity bit zero. Zero traffic with a healthy sensor is a valid zero. Unknown traffic because of a failed sensor is invalid. Store the complete transform vector in the model bundle.

```
node_x:       float32[N, 88]
edge_index:   int64[2, E]
edge_attr:    float32[E, 22]
global_raw:   float32[48]
global_embed: float32[24]
entity_ids:   opaque[N]       # sidecar, never embedded as raw IDs
target_mask:  bool[N, J]
```

11. Targets, Labels and Risk Sets

Targets are a versioned event ontology, not the word 'malicious'. Tier A supports named targets only when their label manifest is qualified: first observed malicious endpoint contact, confirmed initial access, confirmed lateral movement, confirmed C2, and verified exfiltration. The first target is explicitly an observable proxy; it MUST NOT be renamed compromise. Unsupported targets return `not_supported` rather than zero risk.

For each endpoint and event type, an episode has an at-risk interval, a first qualifying event time or right-censoring time, label confidence tier and evidence IDs. Earlier suspicious or malicious behavior is retained when forecasting a later transition. A separate initial-onset track may exclude already-onset entities from its risk set; it must not discard all malicious history for every target.

An episode's event status may be confirmed, weak proxy, synthetic, unresolved or not observed to the end of follow-up. Unresolved ground truth and right-censoring are not synonyms. Unknown attack identity is also not a censoring state. Weak labels may train a separate weak-supervision objective or exploratory task; headline confirmed-event metrics exclude them and disclose the resulting sample count.

ATT&CK tags describe evidence and objectives, not a compulsory linear kill chain. Use multi-label tactic/technique tags with dataset-specific provenance. The SIH projection includes Reconnaissance, Initial Access, Lateral Movement, Command and Control, and Exfiltration; Unknown means insufficient categorization, not an extra malicious tactic. Mere high byte volume never becomes a verified exfiltration label. [23],[24]

Events can co-occur across types, so the default uses separate first-occurrence processes, not a mutually exclusive softmax over all future tactics. A separate first-next-event task may use mutually exclusive marks plus no-event mass, with ties handled by a documented composite/ordering rule. Reset after a qualifying event only when a newly defined episode begins; do not count repeated alerts as independent detections.

12. Probability and ETA Contract

Let `H_d` be the eligible history at decision time d. For endpoint e and event k, `q[e,j,k]` is the probability of the first qualifying event in future bin j, conditional on no earlier occurrence of that event in the episode and on H_d. With 30-second bins, 5/15/30 minutes correspond to 10/30/60 steps. The displayed risk is cumulative, not a risk head with a moving five-minute target reused at each step.

```
S[e,0,k] = 1
p_event[e,j,k] = S[e,j-1,k] * q[e,j,k]
S[e,j,k] = S[e,j-1,k] * (1 - q[e,j,k])
F[e,h,k] = 1 - S[e,h,k]
p_no_event_by_H = S[e,H,k]
```

Use log-survival sums and stable log1p for numerical safety. Cumulative risk MUST be nondecreasing in horizon and lie in [0,1]. Different event-type risks need not sum to one. For a three-model ensemble, average each member's cumulative distribution; do not multiply averaged hazards and call it the same mixture.

ETA is derived from event-time mass. Publish a conditional median or interval **given occurrence within the supported horizon**, together with that occurrence probability. If no-event mass dominates or the distribution is unsupported, withhold the countdown. No independent scalar ETA head is allowed to contradict the event-time distribution.

Default maximum modeled horizon is 30 minutes. Default headline evaluation is five minutes; 15 and 30 minutes become published claims only after horizon-specific acceptance. A shorter trustworthy horizon is preferable to an unsupported long one. A forecast under observed baseline practice is not automatically a no-defense counterfactual: its `policy_context` states the behavior policy represented in training.

13. Selected Reference World Model

The reference implementation is an **entity-conditioned graph-temporal world model**. It keeps the practical GAT/LSTM lineage but removes the global-pooling identity bottleneck. This is a reasoned starting architecture, not an experimentally established winner. Section 31 specifies the tournament that may replace it. TGN and heterogeneous transformers are credible alternatives, not mandatory upgrades. [6],[7],[25]

The observed graph encoder uses three directed GAT layers: 88 -> 128 with four 32-wide concatenated heads; 128 -> 128 with four 32-wide heads; 128 -> 64 with four 64-wide averaged heads. Use ELU, per-graph normalization over valid nodes only, and 0.1 attention dropout. Reverse communication is a separately observed directed edge. Test whether explicit incoming/outgoing channels improve over standard message aggregation. Edge attributes influence attention; this is not assumed to be a rich edge-message model without an ablation. [6]

For each entity, a GRUCell with input/hidden width 64 updates a 64-dimensional memory from its graph embedding. Masked mean and max pooling of entity memories produce 128 dimensions. Concatenate the global adapter's 24 outputs to form 152 inputs to a two-layer LSTM, hidden width 128. The global LSTM state is the full `(h,c)` pair, shape `[2,B,128]`; entity memories are `[B,N,64]` with identity sidecars and validity masks.

Endpoint heads receive `[entity_memory || global_h_top]`, width 192. Future-edge heads receive `[source_memory || target_memory || global_h_top]`, width 256. Raw entity IDs and slot positions never become learned identity shortcuts. Prediction equivariance is tested by renaming/reordering nodes and checking that outputs follow the corresponding entity.

Concrete heads use ELU between linear layers: hazard `192 -> 64 -> J` for J manifest-qualified event types; tactic `192 -> 64 -> 5`; edge `256 -> 64 -> 1`; global adapter `48 -> 48 -> 24`; global observation decoder `128 -> 64 -> 48` for 24 means and 24 log-scales. Hazard, tactic and edge logits use sigmoid at probability readout, not inside BCE-with-logits. The global decoder reads the imagined top-layer global hidden state. `L_obs` is the equal-weight mean of separately normalized node and global NLL terms; unavailable channels are masked, not zero-valued targets.

Tier A uses 20 observed windows (10 minutes) before each forecast. It recomputes state from that same bounded history, starting from a documented zero state, so training and serving have identical context. Ring buffers may cache raw features, but MUST NOT reuse an indefinitely warmed recurrent state with a model trained only on reset clips. Tier B introduces persistent state with explicit burn-in, checkpoint and prequential evaluation rules.

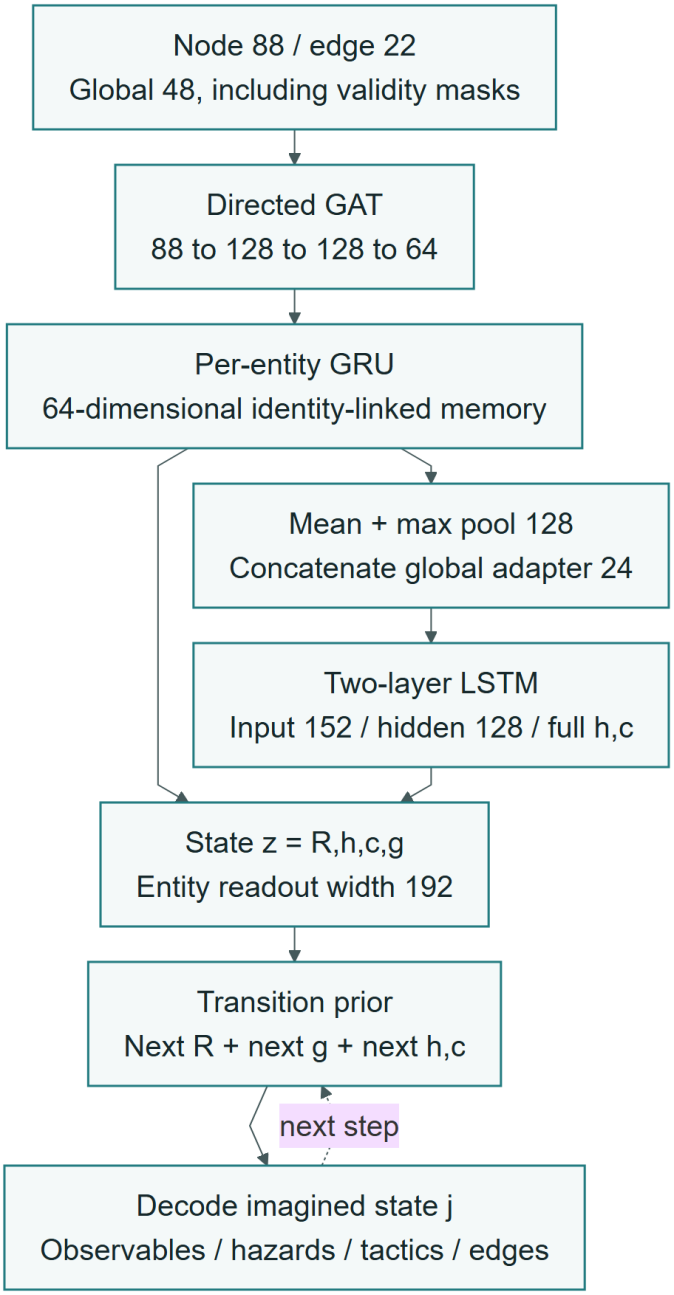


Figure 4. Entity-conditioned graph-temporal reference model. Imagined steps use learned transitions, not future telemetry.

14. Learned Transitions and Observable Rollout

State is `z=(R,h,c,g)`: entity memory R, global LSTM state h/c, and 24-dimensional global context g. The core transition prior is deterministic in latent space and probabilistic in its observable/event emissions. It does not pretend that hand-added Gaussian noise creates calibrated alternate realities.

For each entity, an MLP `192 -> 128 -> 64` predicts its next latent memory from current entity memory and global context. A second MLP `128 -> 64 -> 24` predicts next global context. Pool predicted entity memories and concatenate predicted global context; advance the two-layer LSTM with that 152-dimensional input. This produces the next imagined state without consulting future telemetry.

The observation decoder maps 192 -> 128 -> 88 outputs: means and log-scales for 44 standardized future node values. Bound log-scales to [-5,3]. Use Gaussian NLL in the documented transformed feature space; this is a marginal predictive distribution, not a claim of a fully physically consistent joint packet simulator. Global values have an analogous 24-value decoder; candidate edge existence has a Bernoulli head. Decode future observations at every rollout step and record their error by horizon.

Event hazards and tactic tags are read from each predicted future state, with the bin convention fixed in tests: state after transition j represents bin j. Hazards condition on survival of the corresponding target process. The model is trained using that same convention; no readout is silently shifted by one window. A direct multi-horizon hazard head is a baseline/ablation, not a substitute for the required explicit world-state rollout.

Predicted means are mapped back through the scaler for display; bounded fractions are projected to valid ranges and projection frequency is reported. Excessive projection is a model-health failure. Unknown future entities receive an explicit outside-candidate output; they are not fabricated latent hosts. Tier A edge output is marginal future communication probability within the scope. Joint attack-path distributions require the Tier B/C typed-event model.

The architecture is falsified as a useful world model if future-observation error does not beat persistence, rollout loses calibration, or removing transition/decoder training leaves forecasting utility unchanged within the predeclared margin. Latent next-state error alone is not sufficient evidence. [8],[9]

15. Training Objectives and Schedule

Every loss is normalized by its valid targets, then logged by task, label tier and horizon. Training labels may access future observations; runtime features may not. The training-only target builder and online feature builder are separate modules and processes in the sealed demo.

```
L_survival = -sum(valid_risk_bins * (
    y_event * log(q) + (1-y_event) * log1p(-q)
)) / max(1, count(valid_risk_bins))
L_obs = masked_gaussian_nll(future_values, mean, log_scale)
L_edge = masked_bce(future_edge_exists, edge_logits)
L_stage = masked_multilabel_bce(future_tactic_tags, stage_logits)
L = 1.0*L_survival + 0.5*L_obs + 0.2*L_edge + 0.2*L_stage
```

For an observed first event, risk bins end at that event. For a censored endpoint, bins end at observed follow-up; incomplete later bins contribute no negative labels. For multiple supported event types, apply this separately. The definition does not censor an observed missed attack out of evaluation. The discrete-time survival formulation is established methodology; FORESIGHT's application and constraints require their own validation. [10]

Tier A cutoffs lie on the UTC 30-second grid; each future bin is `(d+(j-1)*30s, d+j*30s]`. Publication time remains separately recorded. An event occurring after the cutoff but before publication cannot count as an advance warning. Future edge labels require verified observer coverage for that pair and bin. Unseen edges during sensor failure are unknown, not negatives. Train on all directed pairs within the capped scope when feasible; sampled negatives require recorded inclusion probabilities and corrected losses. Outside-scope entities have a separate coverage target and are never silently labeled absent.

Start with AdamW, learning rate 0.001, weight decay 0.0001, batch size 16 clips, gradient norm clip 1, and maximum 40 epochs. These are adjustable development defaults, not optimality claims. Use curriculum rollout lengths 1, 3, 10, 30, 60 when validation remains stable. All claimed inference horizons must receive free-rollout training and evaluation. Compare curriculum and fully teacher-forced training; do not assert that scheduled sampling guarantees stability.

Observable decoders anchor representations to measured quantities. An optional latent-consistency term uses a stop-gradient EMA target encoder plus variance monitoring; it is disabled initially. A constant-embedding collapse fixture and embedding-rank/variance telemetry are required. Benign-only pretraining is a challenger, not an obligatory first phase. [9]

Default training preserves natural class prevalence. If episodes are oversampled, record sampling probabilities and correct the loss or recalibrate on natural-prevalence data. Weighted BCE shifts optimum odds: with positive weight w, `q/(1-q)=w*p/(1-p)`. Temperature scaling alone cannot generally remove an intercept shift. Fit an appropriate affine/logistic calibrator or use an unweighted probability objective, and verify the resulting cumulative distribution. Never downsample the raw benign events inside a graph merely to balance labels. [11]

Freeze the representation and dynamics before fitting SENTINEL's reference and forecast calibrators. Model selection uses development data, not the final test. Early stopping maximizes event recall at the fixed alert budget subject to calibration, coverage and latency constraints; median lead time alone rewards detecting only easy cases.

16. Calibration and Trust Contract

Trust is a vector of evidence, not one confidence scalar. Every forecast carries task support, calibration population, sample counts, model disagreement if available, sensor coverage, identity ambiguity, freshness, scope truncation, novelty and published horizon. An ECE computed for a dataset is not the confidence of an individual prediction.

```
TrustContract = {
  status: supported | exploratory | abstained | not_supported,
  task_id, horizon_seconds, calibration_id, calibration_regime,
  calibration_event_count, calibration_episode_count,
  sensor_coverage, identity_quality, max_feature_age_seconds,
  scope_coverage, novelty_diagnostics, epistemic_method,
  interval_kind, conditions, abstain_reasons, valid_until_ns
}
```

Fit calibrators on a temporally later, untouched calibration partition with natural prevalence. A practical hazard calibrator uses a shared affine logit transform plus regularized horizon-group intercepts, fitted with the survival likelihood, so cumulative monotonicity follows from hazards. Sparse target/regime cells back off to a declared pooled calibrator or abstain. Freeze the selective publication policy and check calibration again after selection.

For ensemble uncertainty, independently train complete encoders, transitions and heads. Three members are the initial stretch budget. Head-only diversity under one frozen encoder is not full dynamics uncertainty. Ensemble percentile spread is labeled model spread unless coverage has been validated for the claimed interval. [12]

Conformal prediction and conformal risk control are research options for specific error criteria. Their validity depends on sampling/shift assumptions; neither grants universal coverage under arbitrary adaptive attacks. Use episode-block calibration, report coverage by regime, and explicitly suspend the guarantee outside its validated assumptions. Adaptive query selection and optional stopping require separate evaluation. [13],[15]

17. SENTINEL and Unknown Attacks

SENTINEL publishes unexpectedness, not an automatic probability of attack. The minimum implementation uses feature-group predictive residuals against a frozen qualified reference plus an embedding-distance diagnostic. Candidate energy/OpenMax/one-class methods compete on the same held-out campaigns and benign-drift cases. A score derived from the same encoder is correlated evidence, not an independent sensor. [16],[17],[18]

Each evidence stream records its generating sensor, upstream events, transform and shared-model dependencies. Group duplicated flows, endpoint alerts derived from those flows, and repeated threat-intelligence reports. Fusion may use a calibrated stacking model trained on disjoint development folds with missingness indicators; multiplying likelihoods as if all streams were independent is forbidden unless the conditional-independence model is justified and tested.

Unknown type	Potential useful signal	Earliest defensible claim	Principal blind spot
New malware binary	Provenance, process tree, runtime behavior	Pre-execution only with qualified endpoint/content evidence; otherwise during/after execution	Legitimate signed tooling or unseen behavior not captured
Unseen attack family	Structural/temporal behavior and residuals	Before a later observable stage, when precursor evidence exists	Family may mimic the benign reference
Unknown exploit technique	Runtime/privilege transitions and application evidence	During or after exploitation; possibly before later impact	No detectable precursor or instrumented signal
True zero-day vulnerability exploitation	Exploit consequences, exposure context, endpoint evidence	Bounded resilience in tested scenarios, not universal pre-exploit prevention	Encrypted or silent execution with no observable distinction
Novel sequence of known actions	Long-memory event order, permissions and path context	Before a later action if order changes risk measurably	Sparse history or common administrative workflows
Benign novelty / distribution shift	Software rollout, new service, sensor change	Change detection only	Novelty incorrectly interpreted as maliciousness
Adversarial evasion	Cross-domain inconsistencies and invariant violations	Only when observed differences survive attacker control	Observational equivalence

If benign and malicious situations induce indistinguishable available telemetry, no classifier can reliably distinguish them from that telemetry alone. FORESIGHT must seek another observation, constrain exposure through ordinary policy, or abstain. It can detect useful malicious behavior without knowing a vulnerability when behavior has security-relevant observable consequences; it cannot guarantee knowledge of an invisible exploit.

Unknown-memory records have provisional concepts, source lineage, recurrence, analyst confidence and expiry. Labeling a residual as Unknown does not create a new confirmed attack family. Evaluate false novelty on patch rollouts, backups, employee onboarding, routing changes and sensor upgrades, not only unknown attacks.

18. Operational States and Degradation

Use orthogonal state axes rather than forcing high risk, novelty and broken telemetry into one exclusive enum.

Axis	Values	Meaning
Forecast support	supported, exploratory, abstained, unsupported	Whether the stated task can be published
Threat evidence	low, elevated, confirmed_observation	Assessment under available evidence; not hidden-state certainty
Novelty	reference_like, novel, unavailable	Unexpectedness relative to the reference
Health	healthy, degraded, stale, failed	Input/model/runtime health
Action authority	observe, recommend, approved, executing, blocked	Separate control-plane state

The compatibility UI may display NORMAL/ELEVATED/NOVEL/DEGRADED, but it MUST retain the full axes. Missing data can coexist with elevated risk. High risk does not extend a horizon beyond its calibration. No model score is emitted as a confident zero because data is absent.

The deadline ladder drops optional explanation work first, then optional path search, then unsupported long rollout. A short-horizon forecast is allowed only if independently validated. A static detector fallback is labeled detection, not forecasting. Parser or state-integrity failure produces abstention and an operational alert. Cached results include age and the original decision time; stale results cannot authorize new actions.

Component diagnosis records whether the failure originated in sensors, identity links, feature transforms, representation, transitions, calibration, novelty reference or action model. Automatic diagnosis is initially rule-based; a learned failure-localization model must beat these rules on injected faults before promotion.

19. Explainability and Analyst Workflow

The explanation contract answers: why this entity, why this event, why this horizon, which source observations support it, what is missing, and which additional evidence could change the decision. Preserve feature-group, temporal-window and edge attribution with the exact model state and prefix. Graph attention may be shown as an internal message weight, never labeled causal importance by default. [19]

Grouped deletion/occlusion compares the stated evidence against random and matched-size baselines. Recompute derived statistics after physically meaningful deletions, preserve masks, and flag off-manifold perturbations. Measure comprehensiveness, sufficiency and stability; a negative/suppressive feature is not expected to lower risk when removed. Gradient attribution is accepted only when it explains the same selected endpoint/event/horizon score that appears on screen.

Flow provenance is first-class. The UI lists the source flow/packet ranges that contributed to an aggregate, and distinguishes observed flagged evidence from future suspected behavior. The graph can filter by host, event, novelty, missingness and forecast time. Evidence panels carry ATT&CK mapping tier and explicit unsupported claims.

The optional LLM is a read-only explanation/summarization client over structured evidence. Its text is untrusted; it cannot set risk, mutate ground truth, approve actions or issue unrestricted queries. Summaries must cite evidence IDs, escape source content, and label unsupported interpretations. No cloud LLM dependency exists in Tier A.

20. Attack-Path Engine

Maintain separate observed-event, exposure and hypothesized-future graphs. Exposure records include observation time, expiry, evidence source and confidence. Missing CVE or permission information does not prove a path impossible. Only a verified fresh constraint may remove an edge categorically; uncertain exposure contributes uncertainty rather than zero probability.

Tier B candidate generation starts from observed reachable entities, authorized inventory, prior communications and typed permissions. Include a no-event option and an outside-candidate option. Measure candidate recall before judging a neural path ranker. Hard negative sampling includes previously seen but inactive edges, same-role alternatives and plausible permissions, not only random disconnected nodes. [7],[26]

The next-mark decoder conditions on state and previous generated marks. A mark is `(event_type, source, target, time_bin)` with an allowed-type mask. Teacher forcing predicts the next observed mark in training; free generation feeds predicted marks back. Beam search uses a default width of 8 and depth of 4, with explicit truncated mass. Independent edge marginals are not multiplied and relabeled a path posterior. Event-time hazards and next-mark outputs require consistency checks or are shown as separate tasks.

```
P(path | H_d) = product_l P(mark_l | H_d, mark_1..mark_(l-1))
reported = {ranked_paths, no_event_mass, outside_candidate_mass,
            truncated_search_mass, calibrated: false_or_evidence_id}
```

Top-k path correctness uses a predeclared typed-event equivalence rule and temporal tolerance. Enterprise objective hypotheses such as credential expansion or data access remain latent hypotheses, not actor attribution. Tier A may show a ranked candidate-edge projection; a calibrated multi-stage path claim is reserved for a task with sequence labels and held-out path evaluation. Alert-driven next-action forecasting already exists in research, so this capability alone is not new. [26]

21. Active Perception

Evidence acquisition must improve a decision, not merely decrease entropy. The default stretch policy is a transparent rule: request a missing process tree for ambiguous remote execution, identity/session detail for uncertain credential use, or an allowed higher-fidelity network observation. Requests are allowlisted, scoped, costed, expire, and return unavailable/denied/time-out states without inventing observations.

A learned selector requires a conditional model of the possible answer to each query, trained on genuinely paired modalities. Offline masking of already-collected evidence is a useful prototype but understates live collection delay and selection bias. The query answer must have its own occurrence, availability and receipt times; it produces a new forecast version, never rewrites the old seal.

```
value(q) = min_a E[L(a,Y) | H_d]
           - E_answer[min_a E[L(a,Y) | H_d, answer_q]]
           - collection_cost(q) - privacy_cost(q) - delay_cost(q)
choose q only if value(q) > 0 and policy_allows(q)
```

Waiting and acting immediately are comparison policies. Avoid querying when collection latency exceeds useful intervention time. Test greedy information gain, random allowed query, fixed rules and no-query at equal cost. Learned active acquisition and latent answer models have prior art; FORESIGHT's contribution must lie in measured cyber decision value with timing and safety constraints. [27],[28]

22. Intervention Models and Causal Evidence

The passive forecaster estimates outcomes under represented operating practice. An action model estimates outcomes under a specified intervention. These are separate model artifacts with separate data, calibration and authority. Changing an input feature and observing a score change is sensitivity analysis only. A successful response label requires an independently observed postcondition or outcome, not the model agreeing with itself. [14]

Tier B uses an isolated cyber range with a cloned starting state. Compare no-action and one approved action from the same cutoff, with paired random seeds where possible, varied benign workloads and attacker behaviors, and independently measured service health. A fixed PCAP cannot branch into a different causal future. CybORG/CAGE and CyberBattleSim support research simulations; executable adversary-emulation tooling supports authorized range scenarios, but simulator success is not production proof. [29],[30],[31]

An observational startup action dataset MUST record the policy that selected actions, available alternatives, propensity where known, operator overrides, timing, confounders, outcome follow-up and missingness. Positivity/overlap, consistency, interference and unmeasured confounding must be assessed before off-policy causal estimates. Unsupported actions or contexts receive bounds/abstention, not extrapolated confident benefits.

The planner chooses among no action, seek evidence, escalate, and allowed controls. It minimizes expected residual loss plus disruption under hard policy, scope and service constraints. Optimize across competing paths and unknown branches, not only the single highest-ranked path. A cost-weighted minimum cut is a useful structural baseline; it is not a causal optimizer by itself.

```
a_star = argmin_a worst_case_over_supported_models(
    expected_security_loss(a) + business_cost(a) + delay_cost(a)
)
subject to allowed_scope(a), service_constraints(a),
           evidence_support(a), authority(a), expiry(a)
```

Reinforcement learning and adaptive attacker/defender policies remain research. They must beat no-action, fixed rules and constrained planning while preserving hard safety checks outside the learner. A constrained reward objective alone does not guarantee a production control invariant. [32]

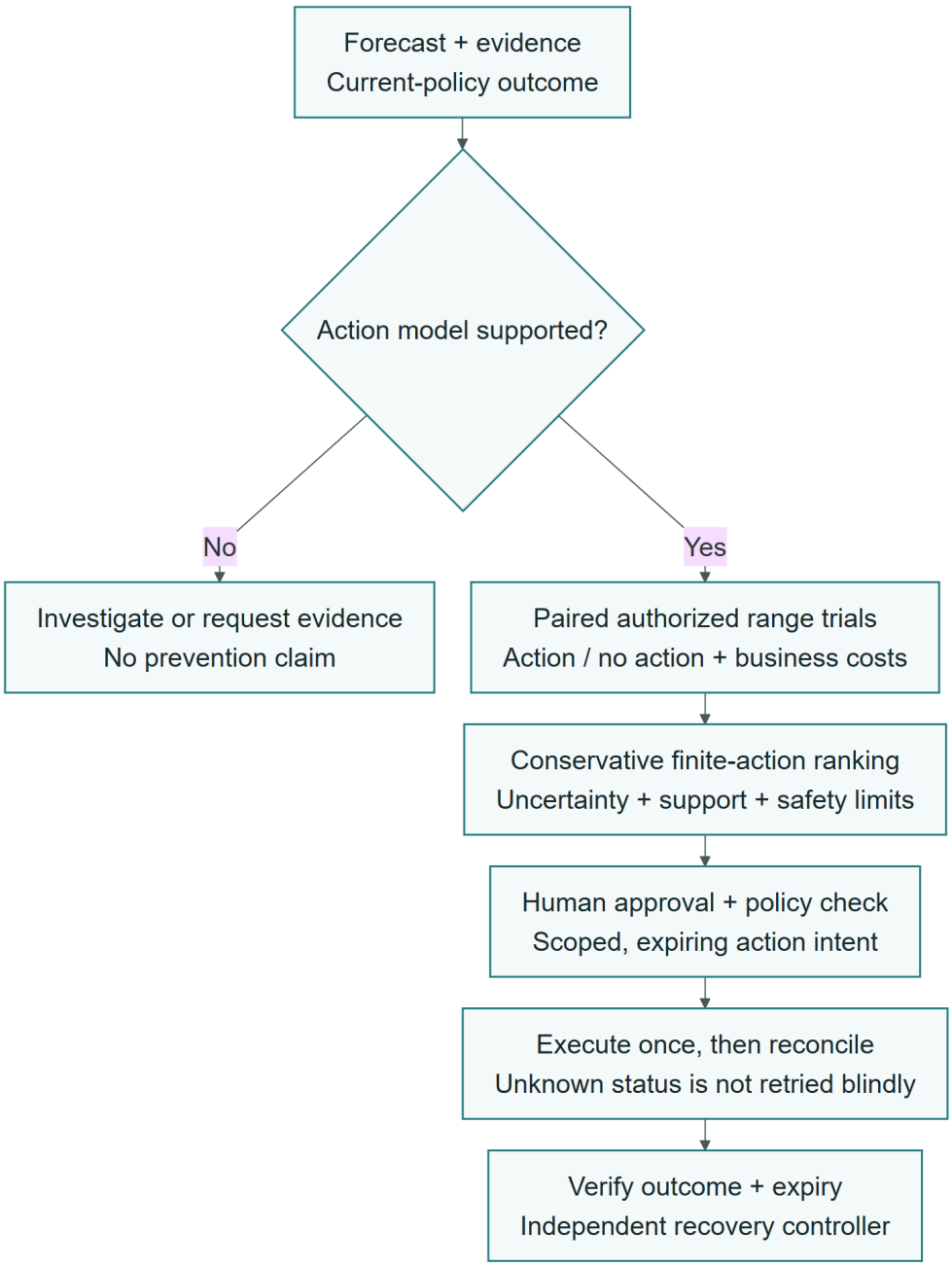


Figure 5. Optional response path. Causal support, authorization and outcome verification are independent gates.

23. Safe Response Execution

All execution passes through an independent policy and approval service. Model processes have no standing administrative credentials. Control requests bind tenant, exact asset identity/version, action type, parameters, maximum scope, evidence/forecast hashes, policy version, expiry and idempotency key. Recheck all bindings immediately before execution to prevent stale approval and identity-race errors.

Action state	Required behavior
proposed	No control effect; include alternatives and uncertainty
authorized	Signed approval bound to the exact proposal and expiry
executing	Durable intent written before connector call
succeeded	Connector acknowledgement plus verified postcondition
failed	Definitive failure; do not claim prevention
unknown	Timeout/ambiguous result; reconcile, never blindly retry a destructive action
expired / recovered	Independent expiry/recovery process verifies the final state

Temporary network restrictions may be reversible at the policy level. Token revocation, process termination, lost sessions and interrupted transactions are not fully reversible. Record `reversibility = reversible | compensatable | irreversible`; use compensating recovery rather than promising rollback of every action. Preservation of security-management connectivity and critical service paths is an explicit range test.

An independent reconciler enforces TTL and recovery even if the planner or UI crashes. High-impact or critical-infrastructure controls require human approval and environment-specific engineering review. No generic automatic isolation policy is authorized for safety-critical systems. Kill switch, dry-run mode, scope allowlist and audit integrity are release blockers.

OpenC2 is an interoperability language, not proof that a connector is safe. ATT&CK/D3FEND mappings explain intent; STIX/TAXII carries structured intelligence; none substitutes for authorization or outcome validation. [23],[24],[33],[34]

24. Continual Learning and Data Moat

The learning pipeline has a frozen long-term reference, a production model, a tenant calibration artifact, a quarantined unknown buffer and a shadow candidate. Active alerts, sensor anomalies, recent containment periods and unverified analyst labels are excluded from automatic normal-reference updates. Adaptation cannot silently make persistent malicious behavior normal.

Candidate training uses approved adjudicated episodes, a retained benign/attack replay set, source-diversity checks and contribution caps by tenant, sensor and campaign. Low-and-slow poisoning may evade alert-based quarantine; random low-risk auditing and delayed adjudication are needed to discover contamination outside flagged periods. Signature, backdoor and adaptive poisoning tests complement ordinary validation. [20]

Promotion requires backward-compatible feature/identity contracts or a migration plan; old and new campaign performance; benign-drift false-alert load; calibration; runtime; and rollback rehearsal. Calibrator changes are also model changes. Shared representation updates cannot consume one tenant's raw data without an explicit permission and retention basis. A fixed global reference prevents a single local adaptation from overwriting shared behavior.

The valuable data unit is a lineage-complete **observation -> forecast -> alternatives -> query/action selection -> actual action -> delayed outcome -> adjudication** trajectory. It includes misses, no-action cases, denied actions, uncertain outcomes and service side effects. A corpus containing only successful interventions is biased and not an efficacy dataset.

Tenant-local raw evidence is the default. Cross-tenant representations can still leak sensitive information; pseudonymization is not anonymization. Differential privacy or federated learning is evaluated against rare-attack utility, membership inference and tenant isolation, not added as a blanket privacy claim. Retention/deletion rules may make exact replay unavailable; retain explicit tombstones and lineage, never claim permanent reproducibility after authorized data destruction.

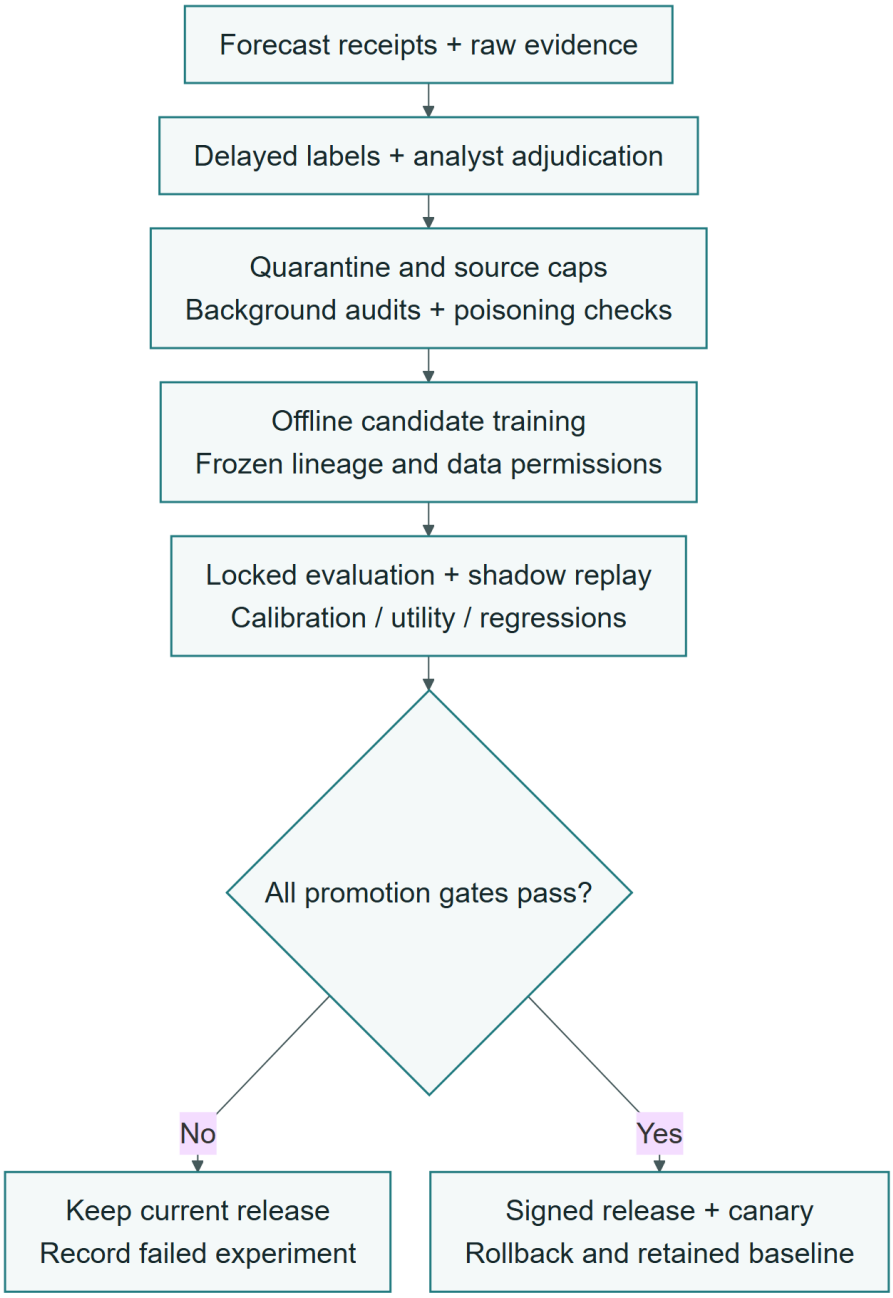


Figure 6. Controlled learning. Candidate models cannot promote themselves or overwrite the retained baseline.

25. Runtime, APIs and Reproducibility

The Tier A stack is Python, PyTorch/PyG, Arrow/Parquet, a native packet extractor, scikit-learn/XGBoost, FastAPI and a local Streamlit or equivalent UI. SQLite stores local run metadata; object files hold immutable evidence. No message broker, cloud API, LLM or Kubernetes cluster is required. Use a tested dependency lock and safe tensor storage; do not load arbitrary pickled checkpoints.

Interface	Input	Output and invariant
POST /v1/sessions	Local uploaded PCAP/CSV + scope	Session and capability report; bounded parsing
POST /v1/forecasts	session, decision cutoff, targets, horizon	New sealed ForecastRecord; no future access
GET /v1/forecasts/{id}	Forecast ID	Immutable forecast and source-linked evidence
POST /v1/queries	Forecast ID + allowed query	QueryProposal; no control authority
POST /v1/replay/reveal	Judge/reveal capability	Suffix and evaluator comparison, separate process
GET /v1/health	None	Component health, versions and backlog
POST /v1/actions/propose	Forecast and policy context	Recommendation only; Tier B/C
POST /v1/actions/{id}/authorize	Exact proposal + approval	Scoped authorization; separate control service
POST /v1/actions/{id}/execute	Authorization + idempotency key	Outcome state; dry-run default

Invalid schema returns a structured 422 error; unsupported targets return a typed unsupported result; scope/authorization failure is 403; stale approval or state conflict is 409; timeout yields an explicit job/unknown status rather than an invented success. Large parse/replay jobs are asynchronous with cancellation and quotas. Every read/write is tenant-scoped in production, including object storage and evidence exports.

```
ForecastRecord = {
  id, schema, tenant_id, scope_id, session_id,
  decision_cutoff_ns, decision_emitted_at_ns, prefix_hash,
  model_hash, scaler_hash, calibrator_hash, policy_context,
  targets: [{entity_id, event_type, label_tier, status,
    bin_seconds, hazards, cumulative_risk, event_time_mass,
    no_event_mass, conditional_eta, trust}],
  future_observables, candidate_edges, paths,
  novelty, evidence_ids, runtime_health, forecast_hash
}
```

The manifest records source/code/environment/model/calibrator hashes, dirty-tree hash, data splits, parser transforms, seeds, hardware, row losses, scope coverage and full commands. A release includes a small legal golden dataset fixture and a larger data-retrieval manifest. Training-data licensing and redistributable weights are reviewed separately from the project's open-source code license.

26. Inference Algorithm

```
def forecast(request, artifacts):
    d = request.decision_cutoff_ns
    prefix = store.read_eligible(request.scope, d)
    validate_clocks_lineage_and_scope(prefix)
    clips = features.build(prefix, windows=20, seconds=30)
    R, h, c = zero_state_for_scope(request.scope)
    for graph, global_raw in clips:
        U = graph_encoder(graph)          # valid nodes only
        R = entity_gru(U, R, graph.mask)
        g = global_adapter(global_raw)    # values + validity
        f = concat(masked_mean(R), masked_max(R), g)
        h, c = global_lstm.step(f, h, c)   # full 2-layer state
    observed_state = freeze(R, h, c, g)
    z = copy_state(observed_state)
    outputs = []
    for j in range(1, request.max_bins + 1):
        z = transition_prior(z)
        outputs.append(decode_observations_edges_hazards_tags(z))
    event_distributions = survival_transform(outputs.hazards)
    calibrated = calibrate_hazards_then_survival(outputs)
    trust = assess_support_coverage_freshness_and_selection(calibrated)
    novelty = sentinel.score(clips, frozen_reference=artifacts.reference)
    evidence = explain_selected_supported_targets(observed_state, request)
    record = assemble_and_hash(prefix, calibrated, trust, novelty, evidence)
    ledger.commit(record)                 # before reveal/publication
    publish(record)
    return record
```

Rollout modifies a copied state only. Observation updates never consume imagined future nodes. Explanation replays the same prefix and selected output, not a later live state. Ensemble members have independent states and seeds. Under deadline pressure, shorten only to a validated horizon or abstain; the trace records the executed path.

27. Production Deployment and Scale

The startup deployment uses tenant-isolated collection, a durable partitioned event log, idempotent normalization, partition-affine entity state, stateless model-serving workers with pinned artifacts, object storage for evidence and a separate control plane. External OCSF compatibility reduces adapter fragmentation without forcing all internal model tensors into an event interchange format. [35]

Each entity has one current writer/lease owner; shard epochs prevent stale writers after failover. Exactly-once effects are not assumed from transport delivery: deduplicate event IDs, checkpoint input offsets with state, and reconcile repeated action intents. Cross-shard context includes a version and age; stale neighbors reduce support. Outage replay catches state up but does not rewrite already issued forecasts.

Backpressure prioritizes critical evidence and control receipts, then bounded summaries, with explicit drop/aggregation accounting. It never silently drops low-volume hosts merely because attackers can keep them quiet. Per-tenant quotas limit event rate, graph expansion, query fanout, explanation cost and retained evidence. Control execution has an independent queue and circuit breaker.

Initial design budgets are a Tier A p95 compute-to-ledger latency of 2 seconds and p99 of 5 seconds at the declared demo scope, with an 8 GiB process-memory ceiling. These exclude the unavoidable time spent collecting a 30-second window and must not be reported as event-to-warning latency. Production budgets are set from a measured workload envelope, not copied from these numbers. Report events/second, entity count, edges, feature fanout, p95/p99 total event-to-decision latency, recovery point, recovery time and storage cost together.

Deployment choices are local/on-prem first, then a customer VPC pilot, then multi-tenant managed operation. Support one proven deployment profile before promising three equivalent products. Collector privilege, key rotation, schema migrations, model rollout and disaster recovery are acceptance work, not diagram labels.

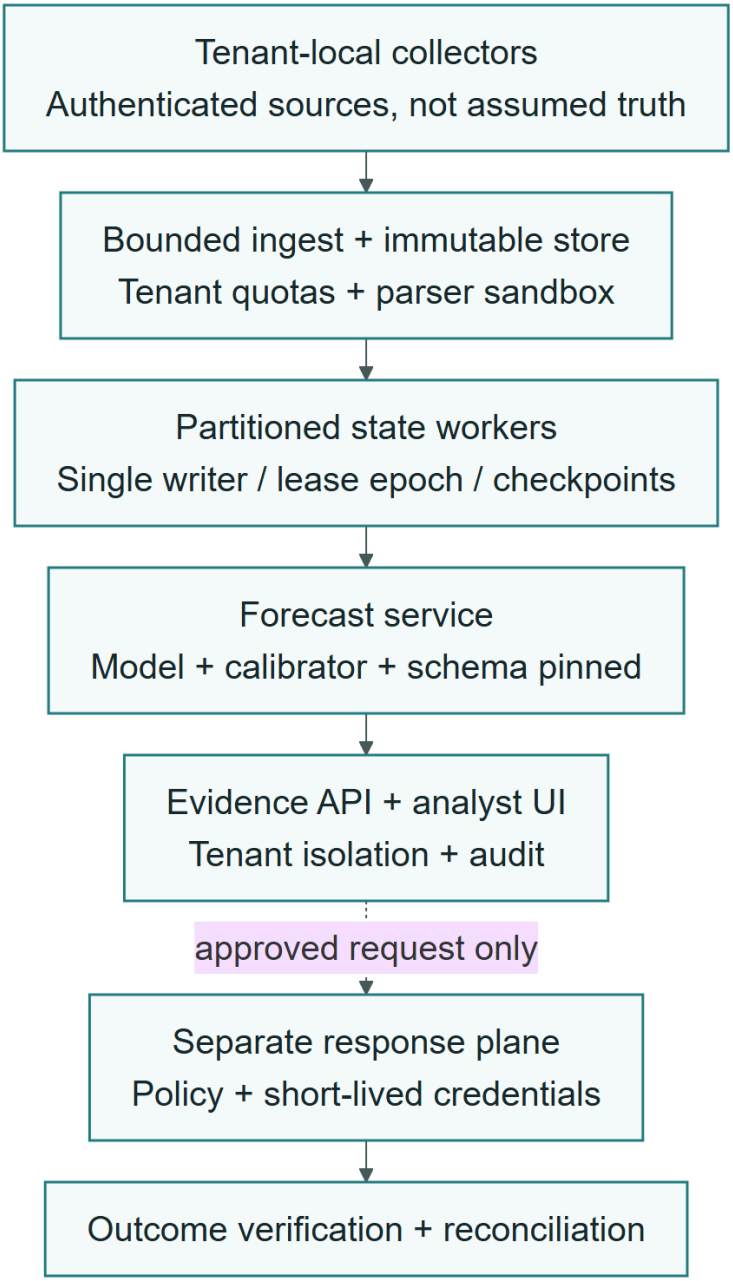


Figure 7. Startup deployment boundaries. The response plane is separate from the forecasting service.

28. Threat Model and Security Controls

Attack surface	Required control	Falsification test
Malformed telemetry / parser exploitation	Sandboxed parser, limits, no shell interpolation, patched dependencies	Corpus fuzzing, decompression bombs, cancellation and crash isolation
Forged/replayed sensor records	Authenticated collector, sequence/replay checks, receipt clocks and provenance	Duplicate, reordered, backdated and validly signed false records
Compromised collector	Trust separation and cross-source conflict reporting	One sensor lies while another disagrees; no confidence inflation
Model/data poisoning	Quarantine, signed lineage, contribution caps, shadow/reference tests	Slowly poisoned normal traffic and targeted backdoor scenarios
Evasion and novelty flooding	Multi-timescale evidence, quotas, aggregation, coverage-aware budgets	Benign mimicry, low-rate progression and sustained alert flooding
Entity resolution abuse	Versioned identity links and approval revalidation	IP reuse, NAT collision, process-ID reuse and stale target mapping
Model extraction / inversion	Tenant access controls, quotas, minimal external score detail	Repeated queries and privacy evaluation within authorized tests
Artifact / supply-chain compromise	Safe model format, signatures, dependency lock, SBOM, restricted CI	Altered checkpoint, unsigned calibrator and malicious archive
Cross-tenant leakage	Tenant keys, row/object isolation and scoped cache/state keys	Same entity IDs across tenants; export/cache and backup restore tests
Response connector abuse	Separate policy, scoped credentials, idempotency and expiry	Stale approval, duplicate execution, timeout and wrong-tenant request
Corrupted evidence	Forecast/content hashes and append-only authenticated ledger	Tamper with prefix, output, policy or source object
LLM prompt injection	Read-only bounded tools and untrusted source text	Telemetry asks summarizer to change risk or execute a command
Insider misuse	Least privilege, dual approval for control changes, audit	Analyst attempts label promotion or unauthorized scope expansion

Authentication is necessary but not proof of truthful telemetry. Cryptographic integrity does not repair wrong ground truth. Security tests follow an explicit attack model and authorization boundary; the product is not released on unit tests alone. NIST's adversarial-ML taxonomy supports the threat categories, but FORESIGHT-specific mitigations still require these tests. [20],[21]

29. Falsification and Benchmark Program

Splits and Independence

Use episode/campaign-level chronological training, development, calibration and locked test partitions. The default target proportions are 60/15/10/15, adjusted only to keep complete episodes and task support. Embargo at least 40 minutes for 10-minute history plus 30-minute forecast wherever raw records could cross boundaries; also exclude complete shared episodes. Stateful challengers must reset or replay only legally available past observations and may not update on future labels.

A family held out from supervised training MUST also be excluded from representation pretraining, threshold tuning and model selection when claiming genuinely unseen-family evaluation. Separate development unknown families from final unknown families. Cross-dataset transfer freezes preprocessing, ontology mapping and thresholds; missing fields are masks and a reported shift, not target-domain fitting. [5],[36]

Common Baselines

Baseline	What it tests
Persistence and rolling empirical event hazard	Whether learned dynamics and hazard add value
Logistic regression on current and legal lagged features	Official comparator and a non-strawman temporal baseline
Gradient-boosted trees with the same lagged features	Strong tabular forecasting opponent
Per-host MLP + GRU, no graph	Whether relational message passing matters
GAT without temporal memory	Whether temporal history matters
Corrected global GAT/LSTM	Whether entity memory adds value over the original design
Direct multi-horizon survival model, no future decoder	Whether world-model training improves the actual forecast
Markov / alert-sequence path model	Whether neural path search improves over simple sequence forecasting
Fixed evidence-request rule and random allowed query	Whether learned query selection earns its cost
No-action and fixed approved-response policy	Whether planning improves security plus service utility

All comparisons use the same legal information and endpoint/event definitions. Equal tuning budgets and hardware are reported, not merely identical seeds. A baseline is not required to collapse for FORESIGHT to count as useful. TGB demonstrates why temporal architecture rankings must be measured rather than assumed. [7]

Metric Definitions

An alert is matched once to a compatible endpoint/event within its horizon, before the observed event. Per endpoint/event, use a 300-second cooldown by default; cooldown is an evaluation policy, not event censoring. Count unmatched alerts only where follow-up is adequate; unresolved outcomes form an explicit unknown-outcome category. Report observed-event recall including misses, conditional lead-time quantiles among successful warnings, and actionable lead time after real review/control delay.

Use PR-AUC, precision, recall, F1, FPR, event recall, false alerts per observed asset-hour, precision at analyst budget, coverage and abstention. For survival tasks report time-dependent Brier with censoring assumptions stated, horizon reliability and event-time likelihood. IPCW is used only when censoring positivity and a justified censoring model hold; otherwise provide complete-follow-up strata and sensitivity analysis. A missed observed attack is not removed by Kaplan-Meier.

World-model metrics are masked observable NLL/error, persistence-relative skill, edge candidate recall, edge discrimination and free-rollout degradation. Unknown metrics include AUROC/AUPR, unknown recall at fixed false-novelty budget, benign-change false novelty and time-to-recognition. Path metrics use top-k typed sequence recall with candidate/search coverage. Query metrics are decision loss, cost, delay and information gain. Response metrics are paired outcome differences, service disruption, erroneous containment and recovery time.

Bootstrap uncertainty at campaign/episode level, not per correlated window. Publish the number of independent units and confidence intervals, and label very small samples exploratory. Report per-family, per-target, per-sensor and per-horizon results. No aggregate can hide a target with zero useful recall.

Stress and Ablation Matrix

Experiment	What would disprove the component	Fallback
Rename/reorder all entity IDs	Forecast changes except corresponding output permutation	Fix identity leakage before any benchmark
Remove graph messages	No material event utility loss	Per-host temporal model
Remove entity memory	Same utility under matched compute	Corrected global reference
Remove observable/dynamics losses	Same or better forecast and rollout utility	Reconsider research claim; retain only validated dynamics task
Increase free-rollout horizon	Error grows while confidence stays narrow	Shorten published horizon
Duplicate one sensor's evidence	Risk rises as if independent confirmation arrived	Lineage-aware fusion
Patch rollout / backup / new service	Unknown alarms exceed allowed workload	Diagnostic-only novelty
Drop packets; delay clocks	Forecast becomes more confident or gains apparent lead	Fix availability and trust policy
NAT, DHCP and host churn	Histories merge or controls target a new owner	Identity abstention and scope recheck
Withhold endpoint answer	Query selector still reports evidence as known	Fix observation boundary
No-action versus action clones	No outcome improvement or excess disruption	Recommendation-only / fixed policy
Action timeout and process crash	Duplicate control or expired restriction persists	Block execution release
Gradual poisoning / backdoor	Reference absorbs attack or unrelated tenant changes	Reject candidate and quarantine lineage
Scale cohort and event rate	Silent truncation or unbounded queues	Partition, rate-limit and declare coverage loss

Acceptance Policy

Hard correctness gates are zero tolerated future-label/suffix access, unauthorized actions or cross-tenant leaks in the defined test suite; exact shape/schema consistency; and hand-computed metric agreement. These are test requirements, not a claim that undiscovered vulnerabilities cannot exist.

The default SIH operating target is no more than one false alert per 100 observed asset-hours, with event recall at least 0.50 on supported primary targets. This is a predeclared design target, not an achieved result and not an enterprise SLA. Promotion requires positive paired primary utility versus the strongest comparable baseline, no material calibration/coverage regression, and passing latency/memory budgets. For a small test set, a wide interval produces 'inconclusive', not a fabricated pass. Profile changes are frozen before final-test access.

Nominal cumulative-risk calibration targets are ECE <= 0.05 with bin counts disclosed, Brier no worse than the strongest baseline, and no horizon published outside its evaluated support. ECE alone is not enough. False-alert upper confidence bounds and analyst workload drive production decisions; the SIH threshold must not be copied unchanged into a 100,000-asset enterprise.

30. Six-Person Build and 120-Second Demo

Owner role	Primary work	Cross-review
Systems lead	Schemas, scope/time contracts, runtime and integration	Evaluation owner approves invariant changes
Data lead	PCAP/CSV pipeline, dataset audit, entity/timeline alignment	Model owner checks target availability
Model lead	Reference encoder/memory, hazards and observable decoder	Dynamics owner checks rollout
Dynamics/trust lead	Training objectives, SENTINEL, calibration and abstention	Evaluation owner checks probability semantics
Evaluation lead	Baselines, labels, sealed test, statistics and adversarial fixtures	Systems lead checks reproducibility
Product/release lead	UI, evidence, diagrams, offline bundle and rehearsal	Data lead checks displayed provenance

Build order is dependency-based: contract fixtures -> qualified episode -> baseline labels/metrics -> graph-temporal model -> free rollout -> calibration/novelty -> explanations/UI -> locked evaluation -> offline release. The critical path is usable event-level ground truth and leakage-safe evaluation, not choosing a larger neural model. The team integrates daily against one golden replay. No member trains on the sealed test to unblock a weak result.

Tier B begins only after Tier A produces a complete evidence bundle. In a constrained hackathon window, allocate most capacity to completing the core and failure-proofing the demo, not simultaneously starting seven research branches. Exact calendar estimates require the actual competition schedule, available hardware and audited data volume; this plan does not invent them.

Judge Demonstration

Time	Action	Verifiable result
0-15 s	Judge chooses a prequalified held-out episode from a manifest	Offline status, supported targets and unseen-family boundary
15-35 s	Show causal prefix and selected endpoint	Flow/packet evidence and missingness; no suffix accessible
35-55 s	Run and commit forecast	Event, cumulative timeline, conditional ETA, trust, source-linked reasons
55-75 s	Reveal future	Success, miss, novelty-only or abstention scored without concealment
75-95 s	Show one fault or one acquired observation	Correct degraded state or a newly sealed revised forecast
95-110 s	If Tier B validated, show precomputed paired range continuations; otherwise show strongest ablation	No-action/action outcome and service check, or honest forecast evidence
110-120 s	Export ledger and locked comparison	Reproducible result and limits

Never acquire a new 'pre-event' observation after revealing the suffix and score it as if it preceded the original forecast. The query branch uses a separately sealed earlier fork with realistic answer latency. The intervention branch starts from the same pre-action snapshot in an executable isolated range; it is not a modified PCAP animation. Long experiments can be summarized by a signed precomputed artifact, labeled as recorded rather than live.

Release package: source repository, setup README, training scripts, safe model weights, model card, locked configuration, dataset/license manifest, golden fixture, full benchmark report, failure cases, a two-page architecture PDF, at most five presentation slides, and a two-minute video. The full handbook stays a team reference. [1]

31. Research Landscape and Architecture Tournament

What Already Exists

The opportunity is not that nobody has combined AI, network context, anomaly detection and containment. Those broad capabilities already exist. The defensible research question is whether FORESIGHT can produce better **availability-correct, calibrated, entity-specific future-event decisions**, and independently demonstrate when its chosen observations or interventions improve outcomes.

Research line	Established capability	What it does not establish for FORESIGHT
TGN / Heterogeneous Graph Transformer	Timed-event memory and typed relational representation	Better attack lead time on this data; causal identification [25],[40]
Temporal Graph Benchmark	Realistic temporal node/link evaluations; simpler models can be strong	That a graph architecture wins all deployment tasks [7]
PlaNet / Dreamer	Learned latent dynamics and planning/imagination in evaluated environments	A validated enterprise cyber simulator or safe control policy [8],[37]
Mamba / neural ODEs	Selective state-space sequence modeling / continuous-depth dynamics	That irregular timestamps justify their cost in SIH [38],[39]
Neural Hawkes	Continuous-time marked event intensities	Reliable latent attacker intent or coverage under missing enterprise events [50]
Discrete survival models	Event-time prediction with censoring-aware objectives	Correct cyber target labels or automatic calibration [10]
Energy / OpenMax / TGN-SVDD	OOD rejection and dynamic-graph anomaly mechanisms	Universal distinction between benign novelty and unknown exploitation [16],[17],[18]
KAIROS	Temporal provenance-graph anomaly detection and investigation	Pre-event causal intervention benefit [41]
Alert-driven attack forecasting	Next-action forecasting from observed alert sequences	That the general idea of predicting attack paths is new [26]
Active acquisition / Learning-To-Measure	Selecting missing features under uncertainty	Live evidence collection without delay, privacy or policy cost [27],[28]
Conformal methods	Risk/coverage control under stated assumptions	Distribution-free security guarantees against arbitrary adaptive shift [13],[15]
Drift localization	Identifying which observations are affected by drift	Automatic diagnosis of every failing pipeline component [46]
CAGE / CyberBattleSim / Caldera	Simulated defense or authorized emulation environments	Transfer of simulator reward to enterprise safety [29],[30],[31]

KAIROS, TGN-SVDD and alert-driven forecasting are especially important nearest neighbors: graph-temporal anomaly learning, provenance context and next-action prediction are not unexplored. FORESIGHT must compare against their applicable mechanisms or reproducible simplified baselines rather than cite them as proof of novelty. The official research literature also warns that security benchmarks can reward unrealistic evaluation choices. [5],[18],[26],[41]

Commercial system	Capability described in official material	Implication for differentiation
Microsoft Defender XDR	Cross-domain incident correlation, attack disruption and documented predictive-shielding actions	Do not claim that proactive multi-sensor containment does not exist [43]
Darktrace Autonomous Response	Autonomous response to anomalous activity across supported domains	Normal-behavior modeling plus response is not a unique category [42]
Wiz	Security Graph context combined with runtime signals and attack-path analysis	Static-versus-live graph integration alone is not an original moat [44]
Cortex XSIAM	Security analytics and automation workflows	A SOC dashboard with playbook automation is not sufficient differentiation [45]

These are vendor-documented capabilities, not independent performance results, architecture disclosures or head-to-head benchmarks. Public material cannot prove that a competitor lacks a hidden implementation of a proposed mechanism. No market-first claim is made.

Candidate Architectures

Candidate	Information advantage	Principal cost/failure	Decision
A0: corrected global GAT/LSTM	Network topology plus window sequence	Pooling loses entity-specific history	Mandatory reference comparator
A1: entity-memory graph-temporal world model	Local history plus global context; observable rollout	More state and losses; may overfit small campaigns	Selected starting implementation, subject to gates
A2: event-driven TGN + event-time heads	Irregular timing and dynamic entity updates	Memory/order leakage, state recovery and event-rate cost	Tier B/C challenger [25]
A3: hierarchical HGT + multiresolution SSM	Typed cross-domain entities and longer context	Large data demands; missing modalities; export/runtime complexity	Startup challenger [38],[40]
A4: recurrent stochastic belief-state model	Multiple hidden-state hypotheses and answer-conditioned queries	Unidentifiable latents, posterior collapse, simulator dependence	Research branch [8],[37]
A5: marked point-process + constrained path grammar	Explicit event time, next mark and typed sequences	Missing-event model and combinatorial marks	Path/event challenger [50]
A6: observation/decision-support mixture	Routes tasks by observability and avoids unsupported outputs	Router can hide weak coverage or overfit	Composition candidate; must expose routing losses

No numerical tournament score is presented because no candidate has been trained for this project. Engineering expectations are hypotheses: A0 is easiest to reproduce, A1 adds entity history at bounded cost, A2 better matches irregular events, A3 targets heterogeneous enterprise context, A4 supports belief/active sensing if identification is adequate, and A5 directly represents event timing. There is no universal winner.

The tournament uses the same audited event manifest, time/scope contract, training and calibration partitions, tuning budget, final targets and release machine. Stage 1 rejects invalid or unsupported candidates; Stage 2

compares event utility at alert budget; Stage 3 checks calibration, coverage and lower-tail performance; Stage 4 measures total latency, state cost and recovery. A more complex candidate must show a paired utility improvement or a new independently validated capability without a material regression. Otherwise keep the simpler design.

Model Families Not Selected by Default

Graph transformers are considered when typed relations and sparse attention add information, not because attention is fashionable. Neural ODE/SDE models are considered for irregular dynamics only when simpler elapsed-time encodings fail. Mixture-of-experts requires enough regimes and data to train a useful router. Contrastive/masked/predictive pretraining is compared against direct supervised transition learning; it cannot see a held-out family during a claimed unseen-family experiment. [38],[39],[40],[47]

32. Research Candidate Register

These ten mechanisms are research proposals, not ten new mandatory services. Each has a nearest-prior-art basis and a concrete falsification test. 'Potential contribution' means worth investigating after a broader novelty search, not a patentability or originality conclusion. Patent claims require a separate professional prior-art and legal assessment; this document does not certify freedom to operate.

R01. Availability-Conditioned Belief Updates

Mechanism: extend the state posterior with per-source age, receipt delay and coverage so it reasons about missing observations instead of treating the observed graph as reality. Keep occurrence dynamics distinct from the observation/collection process. Forecast an event under an explicitly available evidence set, then compare uncertainty and decision value across alternative latency regimes.

Prior art and distinction: TGN supplies timed memory; latent world models supply hidden state; conformal-under-shift work supplies relevant uncertainty limits. The proposed research difference is explicit availability-conditioned cyber forecasting and auditable late-evidence updates, not the invention of belief states. Classification: **novel composition candidate / potentially publishable evaluation mechanism**. [8],[13],[25]

Data/training/compute: paired event-occurrence and receipt logs, artificial delay fixtures plus genuinely delayed captures; train the transition and observation components separately, then jointly under masked likelihood. Initial cost is one metadata encoder and missingness head; full posterior mixtures are Tier C. **Failure:** delays correlate with attacker activity, unknown collector behavior or missing-not-at-random labels. **Experiment:** compare ignoring delays, masking only, and observation-conditioned inference under equal data; measure emitted-time lead, Brier and coverage. **Gate/fallback:** useful improvement under realistic delay without clean-data regression; otherwise keep the strict clock/mask contract. **Value:** Tier A gains correctness; startup gains more reliable degraded operation.

R02. Dependency-Aware Evidence Fusion

Mechanism: represent source lineage as an evidence-dependency graph; collapse duplicates and shared upstream causes before fusion. Learn a bounded, missingness-aware combiner over groups, with sensor conflict exposed rather than averaged away. A hundred copied alerts must not outweigh one independent observation.

Prior art and distinction: provenance graphs, Bayesian dependence modeling and multi-sensor correlation are established. The candidate contribution is a calibrated fusion contract that binds future-event confidence to lineage-aware effective evidence, with adversarial duplication tests. Classification: **adaptation and novel composition candidate**, not a new theory of evidence. [20],[41],[43]

Data/training/compute: multi-source event lineage and known duplicated observations; fit stacking models on disjoint folds and include synthetic duplication/conflict perturbations. Linear-size grouping over evidence edges is the starting cost; avoid a second giant GNN. **Failure:** incomplete lineage, correlated sensors with no documented parent, colluding collectors. **Experiment:** repeat one signal, remove one truly independent modality and corrupt one sensor. Measure confidence inflation and decision loss. **Gate/fallback:** duplication invariance and improved conflict calibration versus naive fusion; otherwise retain rule-based grouping with no probabilistic fusion claim. **Value:** feasible in Tier B; strong platform-security value in pilots.

R03. Decision-Value Evidence Requests

Mechanism: choose the observation expected to change the best allowed defensive decision after accounting for collection delay, privacy and cost. Query only when its expected value exceeds acting now, escalating or waiting. The output includes predicted answer distribution and policy eligibility.

Prior art and distinction: active feature acquisition and information-gain selection are established, including latent-variable and in-context methods. The proposed distinction is an explicit cyber action-loss and time-to-impact objective evaluated with real query latency and denied answers. Classification: **known technique adapted / potential systems contribution**. [27],[28]

Data/training/compute: paired network/process/identity observations plus query availability and response times; train a conditional answer model, then cost-sensitive selection. Evaluate a small allowlisted query set with Monte Carlo answers, bounded to the decision budget. **Failure:** wrong answer model, myopic choice, missing-not-at-random modalities, benefit arriving too late. **Experiment:** no-query, random, fixed rule, information gain and decision-value policies at equal query cost. **Gate/fallback:** lower realized loss or fewer unnecessary controls, not entropy alone; otherwise fixed evidence rules. **Value:** one-query Tier B demonstration; startup investigation efficiency.

R04. Unknown-Branch Probability Accounting

Mechanism: explicitly track probability outside the enumerated entity/path set and mass truncated by search. Split absent candidates from low-scored candidates. A path list cannot become confident merely because the correct future was excluded before ranking.

Prior art and distinction: open-set recognition, temporal link prediction and beam-search truncation are known. The research candidate is a unified candidate-coverage and forecast-risk accounting protocol for enterprise attack paths. Classification: **novel composition candidate / potentially publishable benchmark**. [7],[17],[26]

Data/training/compute: sequences with known future entities, entity churn, incomplete inventory and labels distinguishing unseen candidates; train outside-set/no-event heads and candidate generator jointly only after measuring generator recall. Overhead is small for the unknown bucket, larger for retrieval. **Failure:** an unobserved future has no label, search mass cannot be calibrated, dataset topology makes outside-set prediction trivial. **Experiment:** controlled candidate removal and new-host scenarios with entity-disjoint splits. **Gate/fallback:** valid coverage accounting and improved selective path recall; otherwise ranked hypotheses without probabilities. **Value:** Tier B path credibility; large-enterprise coverage safety.

R05. Identifiability-Gated Intervention Planning

Mechanism: attach an evidence-support region to each state/action pair. The planner uses outcome estimates only where action overlap and outcome follow-up support them; elsewhere it uses bounds, fixed approved rules or abstention. Include uncertainty over alternative attacker continuations and business constraints.

Prior art and distinction: causal identification, off-policy evaluation, robust planning and constrained RL are established. The candidate difference is a cyber control interface that refuses unsupported counterfactuals and exposes the exact missing causal evidence. Classification: **adaptation / potential systems research contribution**. [14],[29],[32]

Data/training/compute: randomized range interventions first; later adjudicated policy/outcome trajectories. Train separate outcome models and support estimators; finite action enumeration precedes RL. **Failure:** hidden confounding, interference, deployment shift, incorrect service cost, overconfident support detection. **Experiment:** remove action coverage and change attacker policy; compare ungated prediction, rules and gated planning. **Gate/fallback:** reduced false-control cost with maintained security benefit; unsupported contexts remain recommendation-only. **Value:** limited Tier B proof; central startup trust boundary.

R06. Residual-Based Failure Localization

Mechanism: monitor residual signatures at sensor, resolver, feature, encoder, transition, calibration and action layers. Use a dependency map and controlled probes to select which component to quarantine instead of globally reducing confidence. Diagnosis never rewrites historical evidence.

Prior art and distinction: drift detection/localization and fault diagnosis exist, including recent conformal localization research. The candidate distinction is localization over a versioned cyber ML pipeline with measured selective fallback. Classification: **adaptation / novel composition candidate**. [20],[46]

Data/training/compute: injected clock, feature, identity, shift and model faults; real outage logs when available. Start with rules; a small classifier over residual/health summaries is the challenger. **Failure:** simultaneous faults, attacker mimicry of benign drift, incorrect diagnosis causing worse coverage. **Experiment:** blinded compound faults; evaluate correct component isolation, time to recovery and retained forecast utility. **Gate/fallback:** beats

rule-based diagnosis without unsafe action enablement; otherwise publish component health only. **Value:** Tier A instrumentation, Tier C learned diagnosis and SRE efficiency.

R07. Security-Precondition Sequence Constraints

Mechanism: represent transitions such as resource access or credential use with typed preconditions and observable consequences. A sequence inconsistent with permissions, process lineage or service role becomes an investigation hypothesis. The constraints guide candidate generation and explanation, not a claim that all attacks follow a fixed grammar.

Prior art and distinction: attack graphs, provenance and ATT&CK/Attack Flow already represent behavior and dependencies. The candidate contribution is measured combination of soft preconditions with uncertain observation coverage and temporal forecasting. Classification: **known technique applied differently / composition candidate**. [23],[24],[26],[41]

Data/training/compute: permission snapshots with age, event sequences and benign administration examples. Train a residual model over constraint features, compare hard versus soft pruning, and use bounded local graph checks. **Failure:** stale policy, legitimate exceptions, stolen credentials satisfying every modeled condition. **Experiment:** valid-admin mimicry, permission drift and unseen sequences. **Gate/fallback:** improved event precision at fixed recall without excluding true paths; otherwise keep constraints as analyst context. **Value:** Tier B small ontology; startup investigation and explainability.

R08. Forecast-Utility Memory Retention

Mechanism: retain a compact entity/campaign memory based on past-only contribution to future-event prediction, with a protected rare-evidence reserve. Compare it with fixed-duration history, recency and simple count summaries. Every eviction has a reason and coverage effect.

Prior art and distinction: recurrent memories, retrieval and state-space compression are established. The candidate difference is a security-utility retention policy evaluated against slow attacks and adversarial memory flooding, not a new memory primitive. Classification: **novel composition candidate / speculative until measured**. [25],[38]

Data/training/compute: long-duration episodes, harmless repetitive background and known delayed progression. Train retention on development episodes only with memory budgets fixed; no future label may guide an online eviction. Cost is a bounded scoring step per memory candidate. **Failure:** destroys rare decisive context, learns dataset timing shortcuts, attacker floods high-scoring records. **Experiment:** equal-byte memory budgets, day-scale gaps and flood stress. **Gate/fallback:** more useful lead/recall at equal memory without lower-tail collapse; otherwise deterministic TTL plus protected evidence. **Value:** not required for SIH; important for long-running enterprise behavior.

R09. Delayed-Adjudication Open-World Learning

Mechanism: unknown clusters move through provisional, investigated and verified states with source contribution limits, delayed outcome maturity and random background audits. A concept can be remembered for investigation without being accepted as benign or used to authorize action.

Prior art and distinction: quarantine, continual learning, replay and analyst labeling are known. The candidate contribution is a lineage/maturity protocol that quantifies adaptation delay versus poisoning exposure and rare-attack forgetting. Classification: **novel composition candidate / potential evaluation contribution**. [18],[20],[41]

Data/training/compute: delayed labels, correlated sensors, recurring benign changes and controlled poisoning periods; maintain a frozen reference and shadow candidate with bounded replay memory. **Failure:** biased analyst labels, malicious behavior never reviewed, coordinated clean-label poisoning. **Experiment:** attacker contamination varying in duration/source diversity, with fixed labeling budget. **Gate/fallback:** improves new-concept utility while preserving prior attacks and reference integrity; otherwise manual dataset curation only. **Value:** Tier B memory UI; Tier C learning system and data quality moat.

R10. Sealed Decision-and-Outcome Trials

Mechanism: commit the observation set, forecast, alternatives, chosen query/action, policy, timing and expected outcome before independently evaluating the future. Include no-action and failed decisions; track action propensity and outcome maturity when meaningful. Bind revisions rather than overwriting previous forecasts.

Prior art and distinction: reproducible experiment logs, prospective evaluation, causal trials and immutable evidence are established. The candidate contribution is a reusable cyber forecasting/intervention benchmark unit joining availability, decision cost and delayed outcome. Classification: **novel composition candidate / strongest near-term publishable artifact**, not a new cryptographic primitive. [5],[14],[29]

Data/training/compute: paired range continuations and real prospective shadow forecasts; no model is needed to implement the ledger. Hashing/storage cost is linear in retained evidence; privacy and retention dominate long-term cost. **Failure:** selective episode inclusion, unobserved outcomes, forged timestamps, hidden test iteration, operational interventions confounding the passive forecast. **Experiment:** independent evaluator attempts to reproduce every result and detect deliberately corrupted records. **Gate/fallback:** prospective audit succeeds and no-action/miss cases remain represented; otherwise do not make efficacy claims. **Value:** Tier A sealed forecast, Tier B trial branch, startup outcome dataset.

Research Priorities

The top five research bets are R10 (prospective evidence protocol), R03 (decision-value observation), R05 (identifiability-gated response), R04 (unknown-branch accounting), and R02 (dependency-aware fusion). They target measurable weaknesses that larger models alone do not solve. R01 and R06 are foundational supporting work. No candidate is described as patentable without a claim-level prior-art search; broad ideas such as belief states, active sensing and attack graphs are already established.

33. Adversarial Review and Resolutions

The following are simulated expert perspectives, not claims that twelve independent people or organizations reviewed the system. Each perspective identifies a strength, weakness, unsupported assumption, failure mode and demanded correction.

Perspective	Strongest / weakest	Unsupported assumption and likely failure	Required resolution
ML researcher	Observable rollout / small campaign data	More losses improve prediction; overfit correlated windows	Task-qualified counts, no-decoder ablation and episode inference
Temporal-graph researcher	Entity memory / streaming semantics	Clip-trained state works indefinitely; memory leakage	Explicit reset/burn-in and predict-before-update protocol
Detection engineer	Multi-source evidence / weak tactic labels	Dataset name equals exploit stage; wrong event truth	Confirmed/proxy ontology and parser capability masks
SOC analyst	Evidence-linked forecast / output overload	More paths are more actionable; alert fatigue	Fixed alert budget, suppression ledger and decision-value queries
Incident-response lead	Separate control plane / reversibility claims	Revocation and termination can be undone; business disruption	Action classes, postconditions and compensating recovery
Adversarial-ML researcher	Frozen reference / adaptive novelty	Quarantine catches all poisoning; slow contamination	Background audits, source caps and long-duration poisoning tests
Distributed-systems architect	Durable event log / shared entity context	One global ordering exists; duplicate state/control effects	Ownership epochs, local ordering and idempotent reconciliation
Production SRE	Health axes / recovery correctness	Retry fixes uncertainty; duplicate action or state divergence	Unknown-result state, replay tests and independent expiry reconciler
Product CISO	Least privilege / evidence truth	mTLS means truthful collector; forged observations	Sensor compromise scenarios and authority separation
SIH/NTRO judge	Real forward simulation / over-scoped demo	A research roadmap counts as running functionality	Tier A traceability, two-page summary and honest 120-second proof
Academic skeptic	Falsification program / novelty claims	Known components imply a new research contribution	Nearest-prior-art comparisons and prospective ablations
Enterprise buyer	Integrates with existing controls / economics	Extra signals pay for themselves; cost and operational burden	Pilot utility, storage/compute cost and service-disruption metrics

Disagreement is resolved by information and authority boundaries: the ML team may propose richer state, but the data owner decides whether targets are observable; the planner may rank an action, but policy authorizes it; the product may want longer horizons, but calibration determines publication; the researcher may prefer a larger model, but the shared benchmark decides promotion. Business-critical service constraints override learned utility.

34. Startup Roadmap and Release Boundaries

Stage	Product scope	Evidence to exit	Explicit non-claim
Prototype	Offline traffic world model and sealed benchmark	Tier A correctness, rollout and utility gates	Enterprise deployment readiness
Design-partner pilot	One customer environment, read-only network plus one endpoint/identity source	Prospective shadow forecasts, analyst feedback, local privacy/security review	General cross-enterprise superiority
Enterprise beta	Multi-source evidence, path/query support, human-confirmed controls	Tenant isolation, reliable adapters, service-cost/outcome evidence, recovery drills	Broad autonomous response safety
Production platform	Measured workload SLOs, secure release/operations, supported control policies	Independent security assessment, long-running fault/poison tests, customer acceptance	Universal zero-day or zero-false-positive guarantee
Research expansion	Belief-state/action models and controlled adaptation	Paired evidence beyond rules and simpler models	Unmeasured categories marketed as capabilities

The initial customer wedge is a forecasting and evidence layer alongside an existing SIEM/NDR/EDR, not a replacement for every security product. Start with a bounded use case such as suspicious credential-driven lateral progression, because outcome definitions, evidence requests and safe response options can be made concrete. This is a product hypothesis to test with design partners, not a market-size forecast.

Costs are measured per observed asset-hour and per million normalized events: collection, feature extraction, state, inference, explanation, retained evidence and analyst time. The pilot must show useful warning and reduced investigation or containment loss without requiring an uneconomic amount of telemetry. A sparse high-value sensor integration may outperform a costly full endpoint product build.

Open-source SIH delivery and startup development are compatible, but they require an explicit licensing plan. The judged source, scripts and redistributable model artifacts remain available as required; tenant evidence stays private. Proprietary value may lie in integrations, operations, adjudicated outcome data and validated service, not in quietly removing required reproducibility. Legal, privacy and patent decisions remain separate qualified reviews.

35. Exact Build Configuration and Golden Contracts

This configuration is the only default reference profile. Development changes generate a new config hash; promoted releases freeze it. There is no legacy alternative configuration in this handbook. A 30-minute model horizon does not authorize publishing an unvalidated 30-minute forecast.

```
schema: foresight.core.2026-09
system:
  name: FORESIGHT
  offline: true
  action_mode: disabled
data:
  window_seconds: 30
  history_windows: 20
  modeled_horizon_bins: 60
  evaluated_horizon_bins: [10, 30, 60]
  primary_horizon_bins: 10
  max_nodes_per_graph: 128
  split_unit: campaign_episode
  split_proportions: [0.60, 0.15, 0.10, 0.15]
  embargo_seconds: 2400
  retain_precursor_attack_history: true
features:
  node_values: 44
  node_validity: 44
  node_input: 88
  edge_values: 11
  edge_validity: 11
  edge_input: 22
  global_values: 24
  global_validity: 24
  global_input: 48
  global_adapter_output: 24
model:
  entity_embedding: 64
  entity_gru_hidden: 64
  global_lstm_input: 152
  global_lstm_hidden: 128
  global_lstm_layers: 2
  endpoint_head_input: 192
  edge_head_input: 256
  latent_noise: disabled
training:
  epochs_max: 40
  batch_clips: 16
  optimizer: AdamW
  learning_rate: 0.001
  weight_decay: 0.0001
  gradient_clip: 1.0
  rollout_curriculum: [1, 3, 10, 30, 60]
  loss: {survival: 1.0, observable: 0.5, edge: 0.2, stage: 0.2}
  seeds: [0, 1, 2, 3, 4]
  benign_pretraining: challenger_only
evaluation:
  alert_cooldown_seconds: 300
  false_alerts_per_100_asset_hours_max: 1.0
  primary_event_recall_min: 0.50
  ece_max: 0.05
  compute_to_ledger_p95_seconds_max: 2.0
  compute_to_ledger_p99_seconds_max: 5.0
  process_memory_gib_max: 8
  final_test_model_selection: prohibited
release:
  measured_results_required: true
```

```
  supported_tasks_from_manifest: true
  auto_promote_research: false
```

Implementation Modules

```
foresight/
  contracts/       # event, feature, target, forecast, trust, action
  ingest/          # pcap, csv, capabilities, lineage, hostile-input worker
  identity/        # entity links, scope registry, cohort/partition policy
  features/        # registry, causal windows, transforms, provenance
  labels/          # training/evaluator-only event and censor construction
  model/           # graph, entity GRU, global LSTM, prior, decoders, losses
  trust/           # calibrator, selection, novelty, component health
  evidence/        # attribution, fidelity, seal and ledger
  runtime/         # bounded inference and immutable replay
  evaluation/      # baselines, event matching, bootstrap, stress, ablation
  queries/         # Tier B allowlisted evidence acquisition
  paths/           # Tier B/C next-mark and path search
  controls/        # separate proposal, policy, executor, reconciler
  learning/        # adjudication, quarantine, reference, shadow, promotion
  app/             # local analyst interface
  tests/           # fixtures, properties, integration, e2e, security
  configs/         # frozen schema, model, task and acceptance profiles
```

Golden Examples

1. Hazards `[0.10,0.20,0.30]` produce cumulative risks `[0.10,0.28,0.496]`, event masses `[0.10,0.18,0.216]` and no-event mass `0.504`. Independent ETA must not disagree with these values. 2. A flow ending after cutoff contributes only qualified partial-prefix observations. Removing its suffix leaves the forecast bytes unchanged. Altering its final bytes or label cannot alter the sealed prefix. 3. Two entities swap tensor positions. Their endpoint predictions swap correspondingly; global output remains unchanged. The root's identity sidecar remains bound correctly. 4. A confirmed target occurs with no prior alert. Event recall counts a miss. The record cannot be converted into a censored success or omitted from the denominator. 5. A sensor fails. Missing fields have validity zero, novelty may be unavailable, and unsupported tasks abstain. No confident benign zero or automatic containment is emitted. 6. The same action request is delivered twice; only one control effect is permitted. A timeout creates `unknown`, reconciliation discovers actual state, and recovery remains active after a planner crash.

The first software milestone implements these examples as tests before model training. A later change to any example requires a reviewed schema/metric decision record, not a silent code fix.

Appendix A. Architecture Autopsy and Decision Record

This appendix records why the source v4 was not a safe literal build contract. The corrections are already integrated into the normative sections above; this is not an overlay. Severity is scoped to the threatened claim or release, not a statement that unbuilt software has an exploitable vulnerability. BLOCKER prevents correct core implementation/evaluation; HIGH threatens a major claim or control boundary; MEDIUM threatens reliability, scale or operational utility; LOW affects presentation or maintainability.

ID / severity / source pages	Finding	Integrated correction
A01 BLOCKER; 7,19,76-80	New contracts and executable-looking legacy examples disagree	One configuration, schema and algorithm; old code is not authoritative
A02 BLOCKER; 7,39	Edge list contains 11 values while corrected shape says 10	Explicit 11 values + 11 masks = 22
A03 BLOCKER; 8,45-46	Endpoint forecasts consume only a global pooled hidden state	Entity memory and endpoint-conditioned shared heads
A04 BLOCKER; 8,33,46,78	New hazards coexist with scalar moving-horizon BCE and independent ETA	First-event hazard likelihood, monotone cumulative probabilities, derived ETA
A05 BLOCKER; 6,38	Allowed lateness can be counted as pre-event warning; flow closure may leak	Actual receipt/emission times; no retroactive feature availability
A06 HIGH; 12,36-39	Dataset names stand in for a verified event/packet training foundation	File-level qualification, event counts, provenance and unsupported task status
A07 HIGH; 30,32-33,76	Global malicious-history discard removes progression evidence	Target-specific risk sets; preserve precursors for later events
A08 HIGH; 46-47	Joint latent MSE can reward an uninformative representation	Observable likelihood, collapse checks; stabilized auxiliary target only if useful
A09 HIGH; 50,80	Novelty fusion maps zero novelty to threat >= 0.5	No novelty-to-probability shortcut
A10 HIGH; 50-51	Gaussian rollout noise/head ensembles are called uncertainty	No arbitrary noise; full independent ensembles and measured interval semantics
A11 HIGH; 46,51,80	Pretraining reference is calibrated before joint model changes	Fit reference/calibration after freezing final artifacts
A12 HIGH; 46,51	Weighted BCE probabilities assumed fixed by temperature scaling	Natural prevalence or corrected weighting and affine survival calibration
A13 HIGH; 58,79	Missed observed attacks are generically called right-censored	Explicit recall/misses and event-time follow-up accounting
A14 HIGH; 47,59-60	Final-test candidate comparisons can become repeated test selection	Development tournament and one frozen final evaluation
A15 HIGH; 47,79	Attack-family predictability is treated as a leakage failure	Test prohibited inputs, identities and timing; keep legitimate behavioral signal
A16 HIGH; 44,55,82	Attention is repeatedly treated as explanation or action evidence	Fidelity-tested, source-linked attribution; sensitivity is not intervention

ID / severity / source pages	Finding	Integrated correction
A17 HIGH; 14,55,57	Feature clamping/replay risks becoming causal response evidence	Separate identified action model; branched range outcomes
A18 HIGH; 54	Ranked edges imply a calibrated joint path	Candidate recall, next-mark dependencies, unknown/truncated mass
A19 HIGH; 53	Multiple model/sensor scores can double-count the same evidence	Lineage grouping and tested dependence-aware fusion
A20 HIGH; 56	Information gain has no answer model or delay consequence	Paired observation model and decision-value comparison
A21 HIGH; 57	Breaking the highest-ranked path ignores alternative futures	Residual loss across supported branches plus hard business constraints
A22 HIGH; 40,67	Revocation/process kill described under blanket reversibility	Reversible/compensatable/irreversible action classes
A23 HIGH; 67,83	Approval, retries and uncertain connector outcomes underspecified	Proposal-bound authorization, expiry, idempotency and reconciliation
A24 HIGH; 49,74	Alert quarantine alone does not prevent slow normality poisoning	Long-term reference, contribution caps, delayed review and background audits
A25 HIGH; 37,74	Signed sensors and timestamps can be mistaken for truthful evidence	Compromised-sensor tests and independent receipt clocks
A26 MEDIUM; 38,75	Top-volume host selection can discard the quiet attacker	Explicit cohorts, rooted neighborhoods, protected targets and coverage accounting
A27 MEDIUM; 42	Slot ordering is called necessary even after invariant pooling	Distinguish identity tracking from permutation invariance; test equivariance
A28 MEDIUM; 45	Bidirectionality is called inherently leaky; fixed model-depth/speed rules overstated	Prefix-bounded alternatives evaluated; chosen recurrent path supports online semantics
A29 HIGH; 50,62	LSTM recurrence omits the complete two-layer hidden/cell state	Typed state and one-step equivalence tests
A30 MEDIUM; 38,73	Dropping malformed rates or downsampling benign traffic distorts graphs	Masks, quality accounting and episode-level sampling
A31 MEDIUM; 41	Encrypted traffic assumes certificate/handshake visibility	Parser capabilities and protocol/sensor-specific support
A32 HIGH; 51-52	Unknowns and attack shapes are described as guaranteed detectable	Observability-bound unknown taxonomy and benign-drift tests
A33 MEDIUM; 52,64	Single global degraded mode hides which subsystem failed	Orthogonal support, novelty, health and authority axes
A34 MEDIUM; 65	Stateless workers plus shared state lack ownership/recovery semantics	Partition ownership, epochs, checkpoints and duplicate handling
A35 MEDIUM; 65,84	Immutable evidence conflicts with deletion; shared embeddings presumed private	Retention-aware replay/tombstones and privacy evaluation

ID / severity / source pages	Finding	Integrated correction
A36 MEDIUM; 68-71	Competing demo lengths and 'seven capabilities' blur core completion	Seven evidence-first core deliverables; stretch only after gates
A37 HIGH; 68	Early baseline slices called fully submittable without required outputs	All official traceability items required for Tier A release
A38 MEDIUM; 23,84	Competition narrative implies existing products only detect/static-map	Vendor-documented overlap and narrower measurable differentiation
A39 LOW; 21-25	Overprinted labels and page-25 diagram overlap undermine readability	Fresh renderer, bounded diagrams and full output visual QA
A40 LOW; 85-86	Reused reference IDs and mixed historical vocabulary obscure authority	Single numbered source register and stable contract identity

Strengths Retained

The core problem framing, offline-first deployment, both traffic levels, traceable predictions, explicit world-model supervision, channel separation, conditional horizons, known failure states, strong baseline ladder, security of the platform itself, scoped controls and outcome learning all remain. The rewrite does not add blockchain, a core LLM, a full malware engine or an autonomous attacker simply to make the diagram larger.

Eighteen-Question Decision Checklist

For every promoted component, its decision record MUST answer: purpose; assumption; assumption failure; attacker manipulation; benign shift; necessity; stronger alternative; measurability; new information; isolating ablation; label/time leakage; train/serve mismatch; open-world support; scale; calibration; safe failure; compute; and defensive utility. The register below binds the main decisions to tests rather than asserting all risks are solved.

Decision	Core assumption / failure	Alternative and ablation	Authority / fallback
Flow-plus-packet graph	Channels are causally available; missing capture breaks features	Flow-only, packet-only and combined tests	Capability masks; task abstention
Fixed windows	Useful dynamics survive aggregation; fast/slow events may not	Event TGN, smaller windows, long summaries	Measured horizon; Tier B replacement
Entity/global memory	Local history adds information; IDs may leak	Global-only and per-host/no-graph tests	Permutation/time fixtures; simpler model
Observable world rollout	State projection improves future utility; collapse/error growth	Persistence and no-decoder survival baseline	Shorter horizon or rejected world-model claim
Hazard event targets	Labels identify measurable first events; proxies may lie	Qualified alternative endpoint targets	Unsupported status, not renamed labels
Novelty evidence	Deviations can help investigation; benign shift overfires	Distance, energy, residual and no-novelty tests	Diagnostic only; no automatic action

Decision	Core assumption / failure	Alternative and ablation	Authority / fallback
Paths	Candidate and sequence support exist; inventory hides true route	Markov/structural/ranked-edge baselines	Hypotheses, unknown mass, no calibrated-path claim
Active queries	Answer model and cost are credible; delay erases benefit	Rules, random, entropy, no query	Escalate/wait; no fabricated evidence
Action planning	Effects supported in tested contexts; confounding/shift	No-action, fixed policy, robust planner	Human review or no action
Adaptation	Approved labels and reference suffice; poisoning persists	Frozen reference and full retrain	Reject candidate; quarantine
Streaming platform	Bounded ownership and recovery; duplicates/stale state	Local reference replay equivalence	Backpressure and explicit degraded scope
Outcome data moat	Outcomes are mature and representative; selective labels bias	Prospective no-action/miss audit	No efficacy claim from biased logs

Appendix B. Sources and Evidence Boundaries

Numbered citations identify primary papers, standards, official dataset descriptions or vendor documentation. Research and product pages were checked during this review; papers are not treated as proof that FORESIGHT achieves their results. Vendor claims are descriptions of offered capabilities, not independent measurements. The source architecture is identified in Section 1. The official problem statement was verified in its expanded portal view, rather than relying on community summaries.

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Evidence boundary: the architecture and research candidates are specifications and hypotheses. No FORESIGHT model was trained, benchmarked or deployed as part of producing this handbook. No arbitrary 'zero-day detection rate', latency result, patentability conclusion or commercial superiority claim is supplied. The team can begin implementing the authoritative contracts; claims are promoted only after their gates pass.